

ESG Rating Disagreement Dynamics and Stock Returns^{*}

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Abstract

Using ESG ratings from four providers for S&P 500 firms between 2010 and 2023, we train neural networks to predict whether ESG rating disagreement will increase over the subsequent year and examine whether these predictions are associated with stock returns. Shapley-based feature importance identifies baseline disagreement as the most informative predictor, consistent with mean-reverting dynamics, while disclosure practices, ESG incidents, and news coverage emerge as secondary predictors. In panel regressions, the predicted probability of disagreement increase is positively and significantly associated with subsequent returns for all four ESG dimensions, whereas baseline levels generally are not, even after controlling for polynomial functions of the baseline level. Corroborating portfolio sorts yield significant High-minus-Low six-factor alphas of up to 47 basis points monthly. Results vary across dimensions and subperiods, with Environmental and Social providing the most stable evidence. Overall, our findings suggest that predicted disagreement dynamics, rather than baseline levels, contain return-relevant information.

Keywords: ESG ratings, Rating disagreement, Machine learning, Sustainable finance

JEL Classification: G12, G14, C45, C53, M14

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1. Introduction

Over the past decade, sustainable investing has grown substantially, with tens of trillions of dollars now managed according to Environmental, Social, and Governance (ESG) criteria (Pástor et al., 2024). This growth reflects both financial and ethical motivations (Starks, 2023; Krueger et al., 2020), with many investors accepting lower expected returns to align portfolios with their values (Riedl and Smeets, 2017; Bauer et al., 2021; Pástor et al., 2021). To identify which firms qualify as sustainable, investors increasingly rely on ESG rating agencies, yet these agencies often provide contradictory assessments of the same firm (Semenova and Hassel, 2015; Chatterji et al., 2016). Unlike credit ratings, where inter-agency correlations typically exceed 0.90 (Cantor and Packer, 1997), ESG rating correlations vary widely because agencies diverge on scope, measurement, and weighting (Berg et al., 2022).

Several studies examine whether ESG rating disagreement levels are associated with stock returns, with mixed findings. Gibson Brandon et al. (2021) find that an equal-weighted long-short portfolio sorted on industry-adjusted ESG rating disagreement generates monthly factor alphas of 23 to 27 basis points. Avramov et al. (2022) provide supporting evidence, showing that high-minus-low uncertainty portfolios earn positive alpha, and further document that the negative relation between ESG ratings and subsequent performance exists only among low-disagreement firms. However, Horky et al. (2024) and Anselmi and Petrella (2025) find that these level effects vanish once standard firm characteristics are controlled. Gibson Brandon et al. (2021) suggest that the premium more likely reflects Knightian uncertainty about ESG information rather than heterogeneous beliefs about firm fundamentals (see, e.g., Miller (1977); Harris and Raviv (1993); Kandel and Pearson (1995); Diether et al. (2002); Atmaz and Basak (2018)), noting that unlike analyst forecasts where investors disagree about the same financial quantity, ESG raters measure different constructs using different methodologies. This interpretation aligns with theoretical work showing that uncertainty about signal precision affects equilibrium prices (Illeditsch, 2011). Consistent with this prediction, Knightian uncertainty is priced in the cross-section of stock returns independently of standard risk factors (Viale et al., 2014). At the micro level, Bachner (2025) provides direct evidence of this ambiguity, documenting that identical ESG rating changes from Bloomberg versus MSCI trigger opposite trading

volume reactions, with abnormal volumes but no abnormal returns.¹

Beyond these return associations, disagreement exhibits cross-sectional variation across firms, with larger and less profitable firms exhibiting higher disagreement (Gibson Brandon et al., 2021), while voluntary ESG reports and third-party assurance are associated with lower disagreement (Kimbrough et al., 2024). Moreover, disagreement responds to changes in the information environment. Christensen et al. (2022) show that mandatory disclosure requirements under the EU Non-Financial Reporting Directive increased disagreement as agencies interpreted newly available data through their different methodologies, though provider-level divergence persists across markets (Anonymous, 2026).

Given that disagreement both varies across firms and evolves over time, theoretical work on ambiguity aversion suggests that these dynamics could matter for returns. Epstein and Schneider (2008) show that when investors cannot assess the precision of signals they receive about an asset, they evaluate outcomes under the worst-case precision, and that expected excess returns decrease with anticipated future information quality. If ESG rating disagreement captures the form of Knightian uncertainty about ESG information argued by Gibson Brandon et al. (2021), then anticipated future information quality could also matter for expected returns.

We investigate this using ESG ratings from four major providers - MSCI, Refinitiv, Sustainalytics, and S&P Global - for S&P 500 firms between 2010 and 2023. As future disagreement changes are not known at portfolio formation, we use multi-layer perceptron classifiers trained with walk-forward cross-validation to predict whether firm-level ESG rating disagreement, measured separately for the Overall ESG, Environmental, Social, and Governance dimensions, will increase over the subsequent fiscal year. The classifiers generate out-of-sample predicted probabilities using only information available at portfolio formation. We then examine whether these predicted probabilities are associated with subsequent stock returns through panel regressions and portfolio sorts, testing whether they contain return-relevant information beyond what baseline disagreement levels capture. Further, to address the black-box critique of neural networks, the analysis employs Shapley Additive Global importanceE (SAGE) values, a Shapley-based framework that decomposes

¹Disagreement also creates broader frictions in information transmission, impeding the timely incorporation of ESG news into prices (Serafeim and Yoon, 2023), increasing analyst forecast errors and dispersion (Liu et al., 2024), reducing stock price informativeness (Hu et al., 2024), increasing crash risk (Sun et al., 2025), and impairing market quality through wider bid-ask spreads (Quotb et al., 2025).

predictive performance into individual feature contributions (Covert et al., 2020; Lundberg and Lee, 2017).

The analysis yields three main findings. First, disagreement changes exhibit out-of-sample predictability over the 2017 to 2023 formation years. SAGE analysis reveals that baseline disagreement level is the most informative predictor of whether disagreement will increase, consistent with mean-reverting dynamics: firms in the highest disagreement quintile are 43 to 60 percentage points less likely to experience subsequent increases than those in the lowest quintile, a pattern that is stable across years and present across industries. Secondary features provide meaningful additional contributions to classification performance, with their composition varying across dimensions. For each of the Environmental and Social dimensions, a disclosure-related variable, disclosure score changes and external audit, respectively, is the most prominent secondary predictor, while news coverage ranks highest among secondary features for the Governance dimension. For Overall ESG, incident counts across all three pillars rank among the most important secondary features after the baseline level. Lagged disagreement changes and pillar-matched ESG incident counts also appear among the top secondary features across all four dimensions, suggesting that recent disagreement trajectories and ESG-relevant events carry additional predictive signal.

Second, we provide evidence that the predicted probability of disagreement increase is positively and significantly associated with subsequent stock returns in the full out-of-sample period, while baseline disagreement levels generally are not. In panel regressions with standard controls and fixed effects, disagreement levels do not reach significance at the 5% level for any dimension. When the predicted probability is included alongside the baseline level, it enters positive and significant for all four ESG dimensions while levels generally do not, and remains significant after orthogonalization with respect to polynomial functions of the baseline level, suggesting that firms facing a higher predicted probability of disagreement are associated with higher subsequent returns. Moreover, logistic models estimated with identical features and walk-forward design as a linear benchmark do not produce consistent return associations, suggesting that the signal depends on nonlinear modeling capacity. Portfolio sorts further corroborate the regression evidence. Under value-weighting, monthly High-minus-Low six-factor alphas are 27 basis points for Environmental, 46 for Social, and 47 for Governance, with Overall ESG marginally significant at 25 basis

points. Under equal-weighting, six-factor alphas are significant for all four dimensions, ranging from 23 to 35 basis points. We further show that the strength of these associations varies across dimensions and subperiods, with the Environmental and Social dimensions providing the most stable evidence across subperiods and also surviving transaction costs under both weighting schemes. Moreover, we find that the predicted probability is not associated with total volatility, idiosyncratic volatility, or market beta for any ESG dimension, suggesting that the return association does not reflect compensation for conventional risk exposure.

Third, interacting the predicted probability with firm characteristics shows that the positive association between the predicted probability of disagreement increase and returns is significantly increasing in firm size and analyst forecast dispersion for three of four dimensions, while baseline disagreement level does not moderate the association for any dimension. Moreover, for the Environmental dimension, the association is significantly decreasing in the firm's Environmental rating level.

Overall, this paper contributes to two strands of literature. First, it extends research on ESG rating disagreement and asset pricing. While prior work offers mixed evidence on whether disagreement levels are associated with returns and finds no predictive power in realized past changes (Gibson Brandon et al., 2021; Avramov et al., 2022; Horky et al., 2024; Anselmi and Petrella, 2025), we provide evidence that the dynamics of ESG rating disagreement contain return-relevant information beyond what baseline disagreement levels capture.

Second, it contributes to the machine learning (ML) in finance literature. Building on advances in ML-based prediction in financial data (see e.g. Gu et al. (2020); Freyberger et al. (2020); Avramov et al. (2023); Chen et al. (2024); Kozak et al. (2020); Chincio et al. (2019)), this paper extends these tools to predict ESG disagreement evolution. While prior work applies ML to predict ESG rating levels (Chowdhury et al., 2023; Bilokha et al., 2024; Del Vitto et al., 2023; Sariyer et al., 2024) or ESG rating changes (Zhu and Rahman, 2024), this is the first study to generate probabilistic forecasts of ESG rating disagreement increases and to examine their association with subsequent stock returns. Moreover, following recent applications of Shapley-based interpretability in finance (De Bock et al., 2024; Berger, 2025), we use SAGE values (Covert et al., 2020) to decompose the classifier's predictive performance into feature-level contributions,

identifying mean-reverting disagreement dynamics and pillar-specific combinations of disclosure practices, ESG incidents, and information environment characteristics as the key drivers of classification performance.

2. Data

Our sample consists of S&P 500 constituent firms, selected to ensure adequate ESG rating coverage across multiple providers while maintaining a homogeneous sample. To align feature availability with return measurement, portfolios are formed each June (formation year t), using ESG ratings from fiscal year $\tau = t - 1$ as features and measuring returns over the July t through June $t + 1$ holding period, where fiscal year τ is labeled by the calendar year in which the firm's fiscal year ends. The resulting formation years in our sample span 2010 to 2023, with return outcomes extending through June 2024.

To identify constituents at each formation date, we obtain historical index constituency data from Wharton Research Data Services (WRDS), using CRSP's S&P 500 index membership file with constituents identified as of the last trading day of June each year.² Following standard asset pricing methodology, we apply a freeze-and-follow approach in which all constituents at formation are tracked through the subsequent twelve-month holding period regardless of index changes. Accordingly, firms exiting the index during the holding period remain in our sample, contributing returns until delisting or the end of the holding period (June of year $t + 1$). Conversely, firms entering the S&P 500 after June are excluded from that year's cohort.

We collect monthly stock returns and security identifiers from the Center for Research in Security Prices (CRSP) and firm fundamentals from Compustat North America, including segment data from Compustat Segments and analyst following and earnings forecasts from the Institutional Brokers' Estimate System (I/B/E/S).³ We further obtain factor

²We construct a security master using the CRSP/Compustat Merged (CCM) database, retaining only high-quality links where $\text{LINKTYPE} \in \{\text{LC}, \text{LU}\}$ and $\text{LINKPRIM} \in \{\text{P}, \text{C}\}$, following standard WRDS practice. When multiple share classes exist for a firm, we select the primary security by prioritizing common stock (CRSP share codes 10 and 11), then by largest market capitalization at formation, then by exchange listing (NYSE over NASDAQ). We map index constituents to our primary CRSP security using PERMCO-level links. We further exclude financial firms (Standard Industrial Classification (SIC) 6000–6999) as they operate under fundamentally different balance-sheet structures and regulatory capital constraints, rendering common accounting ratios and risk characteristics (e.g., leverage and book-based measures) non-comparable across sectors.

³We link I/B/E/S to CRSP using the WRDS ICLINK table, retaining only matches with score ≤ 2 (score 0 = exact CUSIP match, scores 1 and 2 = CUSIP match with minor date or name discrepancies).

returns and the risk-free rate from Kenneth French's data library. The key inputs to our disagreement measures are ESG ratings, which we collect from four providers: MSCI, Refinitiv (now LSEG), S&P Global, and Sustainalytics.⁴ These providers represent the major players in the ESG rating market and feature prominently in academic research on ESG ratings (Berg et al., 2022). To complement these core ESG data, we supplement with disclosure scores from Bloomberg, incident data from RepRisk, and news analytics from RavenPack.⁵ Additionally, institutional ownership, sustainability report indicators, and external assurance indicators are obtained from Refinitiv (LSEG), credit ratings are sourced from Capital IQ, and managerial ability scores are collected from Demerjian et al. (2012).⁶

Ensuring that no future information enters our predictive features requires careful attention to data timing. Following Alves et al. (2025), we assume ESG ratings for fiscal year τ are available to investors by June of year $\tau + 1$.⁷ Similarly, financial statement data are required to have a WRDS public availability date on or before the last trading day of June of year t to ensure the information was accessible to market participants, and analyst data reflect the most recent I/B/E/S summary statistics where the statistical period ends on or before the formation date. For other data sources, we apply analogous timing constraints: Bloomberg disclosure scores use the same timing as ESG ratings, RepRisk incidents and RavenPack news are aggregated over the 12 months ending the last trading day of June of year t , and institutional ownership reflects the most recent 13F filing before the last trading day of June of year t .

⁴MSCI assigns an industry-relative letter rating on a seven-notch scale from *AAA* (highest) to *CCC* (lowest), and also produces underlying numeric component scores on a 0–10 scale at various aggregation levels, the latter being what our data contains (MSCI, 2024). Refinitiv (LSEG) ESG Scores are reported as percentile-rank based values on a 0–100 scale (also available as letter grades from *D–* to *A+*), where higher values indicate stronger relative ESG performance (LSEG, 2024). S&P Global ESG Scores, derived from the Corporate Sustainability Assessment formerly administered by RobecoSAM, are measured on a 0–100 scale, where 100 represents the maximum score (S&P Global, 2025). Sustainalytics reports an ESG Risk Rating that measures a firm's unmanaged ESG risk on a scale starting at 0 (lower is better), grouped into five risk categories: Negligible (0–10), Low (10–20), Medium (20–30), High (30–40), and Severe (40+) (Sustainalytics, 2024).

⁵We match ESG providers, Bloomberg, and Refinitiv ownership data to our panel using ISINs. We extract the six-character issuer identifier (CUSIP6) from the ISIN (positions 3 to 8 for U.S. securities). RepRisk is matched using the nine-digit CUSIP from the primary ISIN. RavenPack is matched using the nine-digit CUSIP in the company names file.

⁶Managerial ability scores are matched directly on GVKEY.

⁷For the majority of S&P 500 firms, fiscal years end in December, and this assumption implies a lag of approximately six months. For firms with earlier fiscal year-ends, the effective lag is longer, making the assumption more conservative.

To document how provider availability evolves over the sample, Table 1 reports ESG rating coverage by provider and formation year. The table shows that Refinitiv (LSEG) and MSCI provide near-complete coverage (93 to 99%) throughout the sample. By contrast, S&P Global enters in 2017 with 88% coverage, and Sustainalytics in 2020 with 91%. This staggered entry affects disagreement measurement: during 2010 to 2016, only two providers are available in our sample, yielding a single pairwise comparison per firm. From 2017, three providers yield up to three pairs, and from 2020, all four providers yield up to six pairs. As a result, the number of firms with at least three raters increases from zero before 2017 to 350 (87%) in 2017 and exceeds 92% from 2020 onward.

Given this coverage structure, we begin out-of-sample testing in 2017, which is the first year when most firms have coverage from at least three providers. This threshold matters because our prediction target is the change in average pairwise disagreement: with only two providers, the target reduces to the change in a single pairwise difference, whereas three or more providers yield an average that is less sensitive to any one provider's methodology.

[Insert Table 1 about here]

To examine how agreement between raters varies across sectors, Figure 1 presents pairwise Spearman correlations between ESG ratings across Fama-French 12 industries.⁸ The figure shows that correlations vary substantially across ESG dimensions. Overall ESG exhibits the highest average correlation (0.33), followed by Environmental (0.20) and Social (0.14), with Governance showing the weakest agreement (0.06). These findings align with Gibson Brandon et al. (2021) and Chatterji et al. (2016), who also report that Governance is the dimension on which rating agencies diverge most.

[Insert Figure 1 about here]

At the provider level, the Refinitiv (LSEG) and S&P Global pair consistently exhibits the strongest correlations across all dimensions (e.g. 0.59 for Overall ESG), while pairs involving Sustainalytics often show weak or negative correlations, particularly with Refinitiv (LSEG). Further, correlations vary meaningfully across industries. For instance, Energy firms

⁸We reverse the Sustainalytics scale before computing correlations so that higher values indicate better ESG performance (lower ESG risk) for all providers.

exhibit above-average agreement across all ESG dimensions (e.g. average correlation of 0.50 for Overall ESG). In contrast, Consumer Durables shows notably weak agreement, with several negative correlations for the Governance dimension.

2.1. Measuring ESG Rating Disagreement

ESG rating agencies differ substantially in their methodologies, coverage universes, and rating scales, making direct comparison of raw scores problematic (Berg et al., 2022). To address this, we compute disagreement metrics following Avramov et al. (2022), transforming ratings into comparable percentile ranks using the empirical cumulative distribution function (ECDF). Because providers use different scales, we first orient each so that higher values indicate better ESG performance (reversing the Sustainalytics scale), then index providers by p and q and proceed as follows.

Specifically, let $\tilde{s}_{i,\tau}^p$ denote the oriented score for firm i from provider p in fiscal year τ . For each provider pair (p, q) in fiscal year τ , we identify the set of firms within our S&P 500 sample that are rated by both providers (the overlap sample) and compute percentile ranks within this common set. Let N_τ^{pq} denote the number of firms in this overlap sample. We assign rank 1 to the lowest (worst) oriented score in the overlap sample, so higher percentile ranks correspond to better ESG performance. The mid-ECDF rank for provider p 's score is then given by

$$\pi_{i,\tau}^{p|pq} = \frac{\text{rank}_{pq}(\tilde{s}_{i,\tau}^p) - 0.5}{N_\tau^{pq}} \quad (1)$$

where $\pi_{i,\tau}^{p|pq}$ denotes the percentile rank of firm i assigned by provider p , computed within the (p, q) overlap sample in fiscal year τ . The superscript $p|pq$ makes explicit that this rank is specific to the (p, q) overlap, so that when provider p appears in a different pair (p, r) , the overlap sample differs and the rank generally differs as well. The operator $\text{rank}_{pq}(\cdot)$ assigns integer ranks from 1 to N_τ^{pq} within this overlap sample, using average ranks for ties, and the subtraction of 0.5 in the numerator produces a mid-ECDF transformation that yields $\pi_{i,\tau}^{p|pq} \in (0, 1)$. We compute $\pi_{i,\tau}^{q|pq}$ analogously for provider q 's score, ranking within the same overlap sample. By computing ranks within each pair's overlap sample rather than across the full rated universe, this approach addresses potential selection bias arising from systematic differences in provider coverage (Avramov et al., 2022; Berg et al., 2022).

Following Avramov et al. (2022), pairwise disagreement between providers p and q for firm i equals the sample standard deviation of the two percentile ranks. For two values,

this simplifies to:

$$d_{i,\tau}^{pq} = \frac{|\pi_{i,\tau}^{p|pq} - \pi_{i,\tau}^{q|pq}|}{\sqrt{2}} \quad (2)$$

When a firm is rated by more than two providers, we compute pairwise disagreement for each unique pair and average.⁹ Let $\mathcal{A}_{i,\tau}$ denote the set of providers rating firm i in fiscal year τ , and let $K_{i,\tau} = |\mathcal{A}_{i,\tau}|$ denote the number of such providers. The resulting number of unique (unordered) pairs is $K_{i,\tau}(K_{i,\tau} - 1)/2$. Firm-level disagreement is then given by

$$D_{i,\tau} = \frac{2}{K_{i,\tau}(K_{i,\tau} - 1)} \sum_{\substack{p < q \\ p, q \in \mathcal{A}_{i,\tau}}} d_{i,\tau}^{pq} \quad (3)$$

where the sum runs over all unique provider pairs covering firm i in fiscal year τ , using any fixed ordering of providers to avoid double-counting (we also require $K_{i,\tau} \geq 2$). We construct separate disagreement measures for Overall ESG and for each pillar (Environmental, Social, Governance).

2.2. Prediction Target

Our prediction target is whether disagreement increases from fiscal year τ to $\tau + 1$. Comparing $D_{i,\tau+1}$ to $D_{i,\tau}$ directly is problematic because the set of providers covering a firm can change between years, producing mechanical shifts in measured disagreement that do not reflect reassessment by existing raters. For example, if a third provider begins covering a firm and happens to agree with one existing provider but not the other, measured disagreement mechanically shifts even though neither original provider changed its assessment. We therefore compute disagreement changes using only provider pairs that rated the firm in both years.¹⁰ Let $\mathcal{S}_{i,\tau}^{\text{bal}}$ denote this “balanced” set of provider pairs i.e., those (p, q) with coverage in both fiscal years τ and $\tau + 1$. We then compute disagreement separately for each year using only these stable pairs:

$$D_{i,\tau}^{\text{bal}} = \frac{1}{|\mathcal{S}_{i,\tau}^{\text{bal}}|} \sum_{(p,q) \in \mathcal{S}_{i,\tau}^{\text{bal}}} d_{i,\tau}^{pq}, \quad D_{i,\tau+1}^{\text{bal}} = \frac{1}{|\mathcal{S}_{i,\tau}^{\text{bal}}|} \sum_{(p,q) \in \mathcal{S}_{i,\tau}^{\text{bal}}} d_{i,\tau+1}^{pq} \quad (4)$$

⁹Because pairwise disagreement $d_{i,\tau}^{pq}$ absorbs the overlap-specific ranks, we suppress the conditioning notation $p|pq$ in all subsequent expressions.

¹⁰Note that this balanced-pair construction does not introduce look-ahead bias. Features at portfolio formation use only fiscal year τ information, specifically disagreement $D_{i,\tau}$ computed from all rater pairs with coverage in fiscal year τ . The balanced target $\Delta D_{i,\tau}$ requires fiscal year $\tau + 1$ ratings but serves only as a training label. The model receives no information about which raters maintain coverage in $\tau + 1$. It learns from the historical relationship between observable features and subsequent disagreement changes.

Because both measures use the same set $\mathcal{S}_{i,\tau}^{\text{bal}}$, the change $\Delta D_{i,\tau} = D_{i,\tau+1}^{\text{bal}} - D_{i,\tau}^{\text{bal}}$ reflects how the same raters revised their relative assessments of firm i , rather than compositional shifts in that firm’s rater coverage.¹¹ The resulting binary target equals one if $\Delta D_{i,\tau} > 0$ and zero otherwise. Moreover, we require $|\mathcal{S}_{i,\tau}^{\text{bal}}| \geq 1$.

To summarize the timing convention, features use $D_{i,\tau}$ computed from Equation 3 using all available pairs in fiscal year τ , which is observable at portfolio formation in June of year $t = \tau + 1$. The target uses the balanced measure from Equation 4, which requires fiscal year $\tau + 1$ ratings and is thus not observable until approximately June of year $t + 1$. This separation ensures no look-ahead bias enters our predictive features.

2.3. Predictive Features

We construct predictive features drawing on prior research on ESG rating disagreement and firm information environments, with detailed definitions and sources for all variables provided in Table A.1 in the Appendix.

We begin with features capturing the disagreement level and its recent dynamics. We include the baseline disagreement level $D_{i,\tau}$ as well as the lagged balanced change $\Delta D_{i,\tau-1}$, computed analogously to Equation 4 using provider pairs observed in both $\tau - 1$ and τ , to capture momentum or reversal patterns in rating dynamics. Next, we include the average ESG rating level and its lagged change, as Christensen et al. (2022) and Kimbrough et al. (2024) document that firms with lower ESG performance exhibit higher disagreement.¹² For ESG dimension-specific models, we replace these ESG measures with the corresponding dimension’s disagreement level, lagged disagreement change, rating level, and lagged rating change.

A second set of features captures disclosure and reporting practices. ESG rating agencies observe the same firm disclosures but apply different methodologies for scope, measurement, and weighting (Berg et al., 2022; Chatterji et al., 2016). Using mandatory disclosure shocks, Christensen et al. (2022) find that increases in ESG disclosure lead to increases in disagreement, as expanded disclosure provides more material for divergent

¹¹Because percentile ranks are computed within each year’s overlap universe, $\Delta D_{i,\tau}$ can also reflect changes in the overlap universe composition for other firms. Our balanced-pair construction removes the mechanical effect from rater entry or exit for firm i itself.

¹²For each provider p , we compute firm i ’s percentile rank within that provider’s S&P 500 rated universe using the mid-ECDF transformation analogous to Equation 1, then average across the $K_{i,\tau}$ providers in $\mathcal{A}_{i,\tau}$.

interpretation. Conversely, Kimbrough et al. (2024) find that third-party attestation and standardized reporting frameworks are associated with lower disagreement by constraining interpretation. Given these findings, we include Bloomberg ESG Disclosure Scores and their year-over-year change, as well as indicators for standalone sustainability reports and external assurance from Refinitiv (LSEG).

Next, we include year-over-year changes in ESG-related incident counts from RepRisk, as Glossner (2021) provides direct evidence that firms with high ESG incident rates exhibit significantly higher rating disagreement across providers. We obtain Environmental, Social, and Governance incident counts from RepRisk and compute the change in each category relative to the prior year. As rating agencies lack shared norms for evaluating ESG outcomes (Christensen et al., 2022), providers may respond to the same incident differently, leading disagreement to shift in the subsequent period.

A third set of features captures firm fundamentals and the broader information environment. Gibson Brandon et al. (2021) document that larger firms, less profitable firms, and firms without credit ratings exhibit higher ESG rating disagreement. Kimbrough et al. (2024) and Christensen et al. (2022) further show that analyst following, news coverage, analyst forecast dispersion, and institutional ownership are also associated with disagreement levels. These characteristics relate to firm complexity and information environment quality, and firms with greater complexity or less external scrutiny may be more likely to experience disagreement increases. To capture these dimensions, we include firm size, profitability, asset tangibility, business segment counts, idiosyncratic volatility, managerial ability (Demerjian et al., 2012), and a credit rating indicator. On the information-environment side, we include analyst following, analyst forecast dispersion, news coverage, news sentiment, and institutional ownership. Industry membership is accounted for through Fama-French 12 classifications to address sector-level differences in ESG materiality (Khan et al., 2016). Finally, we include the number of ESG raters covering each firm ($K_{i,\tau}$) and the lagged change $\Delta K_{i,\tau-1} = K_{i,\tau} - K_{i,\tau-1}$ as disagreement dynamics may differ across coverage configurations, and gaining or losing a rater may reflect shifts in the firm's ESG salience that also prompt reassessment by continuing raters.

2.4. Summary Statistics

The full S&P 500 panel comprises 5,701 firm-year observations with at least one ESG rating across formation years 2010 to 2023. Of these, 5,370 have coverage from at least two

providers, which is the minimum required to compute disagreement measures. Table A.2 reports Spearman correlations among all variables and Table A.3 reports summary statistics for the full panel of 5,701 firm-years. The average Overall ESG disagreement is 0.18, with substantial cross-sectional variation (standard deviation 0.12). Across dimensions, disagreement is highest for the Governance pillar (mean 0.22) and lowest for Overall ESG (0.18), with Environmental (0.20) and Social (0.21) falling between. This ordering is consistent with Gibson Brandon et al. (2021), who find that Governance criteria vary most across providers. Furthermore, disagreement changes exhibit near-zero means but meaningful variation (standard deviation 0.10 to 0.15), with approximately symmetric distributions.

The binary classification targets are approximately balanced, with 51% of firm-years experiencing an increase in Overall ESG disagreement and similar proportions across all ESG dimensions (49 to 51%). This balance supports the use of standard classification metrics and suggests that absent predictive information, disagreement increases or decreases with roughly equal probability.

To examine how disagreement evolves over the sample period, Figure 2 plots disagreement over time by fiscal year, showing that equal-weighted (EW) mean Overall ESG disagreement increases from 0.16 in 2010 to 0.20 in 2023. Figure A.1 shows that this upward trend is present across most Fama-French 12 industries. Nevertheless, Figure 3 shows that the proportion of firms experiencing disagreement increases versus decreases generally remains stable around 50% throughout the sample.

[Insert Figure 2 about here]

[Insert Figure 3 about here]

3. Methodology

Our empirical strategy proceeds in two stages. First, we train classifiers to predict whether ESG rating disagreement will increase over the subsequent fiscal year. Training uses walk-forward cross-validation on an expanding window: the initial training sample spans formation years 2010 to 2016, and out-of-sample predictions are generated for formation years 2017 to 2023. To understand what drives these predictions, we apply Shapley-based interpretability analysis. Second, we examine whether the predicted probability of

disagreement increase is associated with subsequent stock returns using panel regressions and portfolio sorts.

Throughout this section, we use the following index conventions: t denotes the formation year (the last trading day of June in calendar year t), $\tau = t - 1$ denotes the fiscal year used for features, i indexes firms, and $m = 1, \dots, 12$ indexes holding-period months (July of year t through June of year $t + 1$, so that $m = 1$ corresponds to July of year t and $m = 12$ to June of year $t + 1$). In the portfolio analysis, ℓ indexes calendar months in the pooled test-period time series (July 2017 through June 2024).

3.1. Machine Learning Framework

For each firm i at formation year t , let $\mathbf{x}_{i,t}$ denote the vector of features observable at portfolio formation (the last trading day of June of year t), and let $y_{i,t} \in \{0, 1\}$ indicate whether disagreement increased:

$$y_{i,t} = \mathbf{1}\{\Delta D_{i,\tau} > 0\}, \quad \tau = t - 1 \quad (5)$$

As this target requires fiscal year $\tau + 1 = t$ ratings, it is not observable at the formation date. To prevent look-ahead bias, we adopt an expanding-window design in which, for each test year t in the out-of-sample period, we train on all formation years from 2010 through $t - 1$. Because labels through formation year $t - 1$ are observable at the June t formation date, this window uses all available information without incorporating future data.

Formally, let $\mathcal{T}_t$ denote the pooled training set of firm-formation-year observations (i, s) from formation years $s = 2010, \dots, t - 1$, where s indexes the formation year of each observation, and let $n_t = |\mathcal{T}_t|$ denote its size. Using this training set, we estimate a classifier $f_t : \mathbb{R}^d \rightarrow [0, 1]$, where d is the number of preprocessed features, that outputs the predicted probability of disagreement increase by minimizing regularized cross-entropy loss:

$$\hat{f}_t = \arg \min_f \left\{ \frac{1}{n_t} \sum_{(i,s) \in \mathcal{T}_t} \mathcal{L}_{CE}(y_{i,s}, f(\mathbf{x}_{i,s})) + \lambda_t \Omega(f) \right\} \quad (6)$$

where $\hat{f}_t$ is the classifier that minimizes regularized cross-entropy over $\mathcal{T}_t$. Here $\mathcal{L}_{CE}(y, \hat{p}) = -y \log(\hat{p}) - (1 - y) \log(1 - \hat{p})$ is the binary cross-entropy loss, $\Omega(f) = \|\boldsymbol{\theta}\|_2^2$ is an L2 weight decay penalty on all model parameters $\boldsymbol{\theta}$, and λ_t is the regularization strength selected

via cross-validation within the training window for year t . For each test-year observation, the fitted classifier then produces the predicted probability $\hat{p}_{i,t} = \hat{f}_t(\mathbf{x}_{i,t})$.

3.1.1. Model Architecture

We employ multi-layer perceptrons (MLPs), feedforward neural networks suited to tabular classification. Specifically, an MLP with L hidden layers maps inputs through successive transformations:¹³

$$\mathbf{h}^{(l)} = g\left(\mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}\right), \quad l = 1, \dots, L \quad (7)$$

where $\mathbf{h}^{(0)} = \mathbf{x}$ is the input, $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are learnable weight matrices and bias vectors, and $g(\cdot)$ is a nonlinear activation function. The output layer applies a sigmoid transformation to produce a probability:

$$f(\mathbf{x}) = \sigma\left(\mathbf{w}^{(L+1)\top}\mathbf{h}^{(L)} + b^{(L+1)}\right) \quad (8)$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function and $b^{(L+1)} \in \mathbb{R}$ is the scalar output bias.

3.1.2. Hyperparameter Selection

We select hyperparameters via walk-forward cross-validation within each training window, respecting temporal ordering. Specifically, for test year t , we partition the training data (formation years 2010 through $t - 1$) into an initial block (2010 to 2012) and subsequent validation years. The procedure trains on the initial block, validates on 2013, then expands training to include 2013 and validates on 2014, and so forth through year $t - 1$. This yields $t - 2013$ validation folds, each preserving the temporal structure of the prediction problem.

Within each of these validation folds, preprocessing transformations are fitted using only that fold's training years and applied to that fold's validation year. To select among candidate configurations, we compute the average cross-entropy loss across validation folds and apply the one-standard-error rule (Hastie et al., 2015), selecting from among configurations with mean validation loss within one standard error of the minimum the

¹³We focus on MLPs for parsimony. Gu et al. (2020) find that tree-based methods and neural networks both outperform linear approaches for return prediction, but that pairwise comparisons between nonlinear methods are generally statistically indistinguishable.

most regularized option (largest λ , then smallest architecture). This guards against overfitting to the validation sample.

Once the best configuration is identified, we fit all preprocessing transformations on the full 2010 to $t - 1$ window and train the final model for test year t . The specific search distributions governing this procedure are detailed in Table A.4.

3.1.3. Feature Preprocessing and Evaluation

At each formation year t , preprocessing uses only the training sample $\{2010, \dots, t - 1\}$ to estimate transformation parameters, which are then applied to both training and test observations. Continuous features are first winsorized at the 1st and 99th percentiles computed from the pooled training sample. Missing values are then imputed with training-sample medians following Gu et al. (2020), and the resulting features are standardized to zero mean and unit variance using training-sample moments. For categorical variables, industry indicators are one-hot encoded using categories observed in the training data.

To evaluate the resulting classifiers, we use the area under the Receiver Operating Characteristic (ROC) curve (AUC), which measures discriminative ability, where a value of 0.5 indicates random guessing and 1.0 indicates perfect separation. As a linear benchmark, we estimate a logistic model with identical features, preprocessing, and walk-forward design. Classification and return results for both models are compared in Section 4.2.3.

3.1.4. Interpretability Analysis

To identify which features drive predictions, we employ Shapley Additive Global Importance (SAGE) values (Covert et al., 2020). SAGE extends game-theoretic Shapley values to machine learning, treating features as players whose “payoff” is the reduction in prediction loss (here, cross-entropy) relative to a no-feature baseline. In particular, each feature’s SAGE value represents its average marginal contribution to predictive performance across all possible feature subsets. Moreover, unlike permutation importance, which can be misleading when features are correlated, SAGE properly accounts for feature interactions by averaging over all coalitions (Covert et al., 2020). In practice, these values are computed via Monte Carlo sampling and sum to the total loss reduction achieved by the model.

We apply SAGE separately for each ESG dimension (Environmental, Social, Governance, Overall ESG), predicting whether dimension-specific disagreement increases

using dimension-aligned features. For each model and formation year, we then compute SAGE values on the out-of-sample test observations and average across years, following Zografopoulos et al. (2025). This yields dimension-specific feature importance rankings.

3.2. Panel Regressions

We estimate pooled panel regressions relating the predicted probability of disagreement increase to subsequent stock returns, following Zhu and Rahman (2024). Specifically, for each firm i and formation year t , we compute the buy-and-hold return over the twelve-month holding period from July of year t through June of year $t + 1$. To construct this return, let $R_{i,t,m}$ denote the delisting-adjusted return for firm i in holding-period month m of formation year t , and let $m_{i,t}^*$ denote the last month with an observed return.¹⁴ When a firm delists before the end of the holding period, proceeds are reinvested at the risk-free rate $R_{t,m}^f$ for the remaining months, following Mitchell et al. (2002) and Langlois (2023):

$$r_{i,t} = \prod_{m=1}^{12} (1 + \tilde{R}_{i,t,m}) - 1, \quad \text{where } \tilde{R}_{i,t,m} = \begin{cases} R_{i,t,m} & \text{if } m \leq m_{i,t}^* \\ R_{t,m}^f & \text{if } m > m_{i,t}^* \end{cases} \quad (9)$$

Here $r_{i,t}$ denotes the buy-and-hold return for firm i over the twelve-month holding period, and $\tilde{R}_{i,t,m}$ equals the observed return when available and the risk-free rate thereafter.

In a next step, we regress these buy-and-hold returns on the predicted probability of disagreement increase:

$$r_{i,t} = \alpha + \delta \cdot \hat{p}_{i,t} + \boldsymbol{\gamma}^\top \mathbf{Z}_{i,t} + \phi_{j(i)} + \mu_t + \varepsilon_{i,t} \quad (10)$$

where $r_{i,t}$ is defined in Equation 9, $\hat{p}_{i,t}$ is the predicted probability of disagreement increase, $\mathbf{Z}_{i,t}$ is a vector of control variables, $\phi_{j(i)}$ are Fama-French 12 industry fixed effects, and μ_t are formation-year fixed effects.¹⁵ To account for within-firm dependence, standard errors

¹⁴Following Shumway (1997), when both the monthly return (RET) and delisting return (DLRET) are observed, we compound them: $1 + R_{i,t,m} = (1 + \text{RET})(1 + \text{DLRET})$. When RET is missing but DLRET is observed, we set $R_{i,t,m} = \text{DLRET}$. When RET is observed but DLRET is missing (the most common case), we set $R_{i,t,m} = \text{RET}$. When DLRET is missing for a delisting month, we impute based on delisting reason: for performance-related delistings (CRSP codes 500 and 520 to 584) we impose -30% , and for liquidations (codes 400 to 490) we impose -100% .

¹⁵Control variables, following Gibson Brandon et al. (2021), include log market capitalization, log book-to-market, gross profitability, momentum (cumulative return over the 11 months ending in May of year t , skipping the month immediately prior to formation), beta, volatility, analyst forecast dispersion, disagreement level $D_{i,\tau}$, and average ESG rating.

are clustered at the firm level following Cameron and Miller (2015).¹⁶

3.3. Portfolio Construction

At each formation date t , we sort firms into terciles based on predicted probabilities $\hat{p}_{i,t}$, where tercile 1 contains the lowest probabilities and tercile 3 the highest. Portfolios are held from July of year t through June of year $t + 1$. When firms delist, their final return is included for that month, and they exit thereafter.¹⁷

From these tercile portfolios, we compute monthly value-weighted (VW) and EW excess returns. For each formation year t , holding-period month m , and tercile k , the portfolio excess return is:¹⁸

$$R_{k,t,m}^{\text{ptf}} = \sum_{i \in \mathcal{P}_{k,t,m}} w_{i,t,m} (R_{i,t,m} - R_{t,m}^f) \quad (11)$$

where $R_{k,t,m}^{\text{ptf}}$ denotes the excess return on tercile k in holding-period month m of formation year t , and $\mathcal{P}_{k,t,m}$ is the set of surviving constituents in that tercile. Because portfolios are rebalanced annually each June and holding periods do not overlap, each pair (t, m) maps to a unique calendar month. We therefore let ℓ index these calendar months in the pooled test period (July 2017 through June 2024). The H–L spread is then defined as $R_{\text{HL},\ell} = R_{3,\ell}^{\text{ptf}} - R_{1,\ell}^{\text{ptf}}$, where $R_{k,\ell}^{\text{ptf}}$ denotes the tercile k excess return in calendar month ℓ .

To evaluate risk-adjusted performance, we estimate the following time-series regression on the pooled calendar-month series:

$$R_{k,\ell}^{\text{ptf}} = \alpha_k + \beta_k^\top \mathbf{F}_\ell + \varepsilon_{k,\ell} \quad (12)$$

for each tercile and the spread portfolio, where α_k is the abnormal return and $\mathbf{F}_\ell$ contains the Fama and French (2015) five-factor model augmented with momentum (henceforth called FF6).

¹⁶With only seven test years (2017 to 2023), year-level clustering would be unreliable given the small number of clusters (MacKinnon et al., 2023; Petersen, 2009).

¹⁷This differs from the panel specification in Equation 9. Here, delisting proceeds are implicitly redistributed among surviving constituents via weight renormalization, whereas the panel reinvests at the risk-free rate.

¹⁸For VW portfolios, $w_{i,t,m}$ is proportional to end-of-prior-month market capitalization. For July ($m = 1$), we use formation-date market capitalization following Beaver et al. (2007). For EW portfolios, $w_{i,t,m} = 1/n_{k,t,m}$ where $n_{k,t,m}$ is the number of surviving firms in tercile k . Weights are renormalized monthly among firms with non-missing returns.

4. Results

We present our findings in four parts. First, we evaluate out-of-sample prediction performance and examine which features drive predictions. Second, we assess whether the predicted probability of disagreement increase is associated with subsequent stock returns using panel regressions and portfolio sorts, including robustness checks across subperiods, net-of-cost specifications, and model-choice sensitivity. Third, we explore whether this return association varies with firm size, ESG ratings, baseline disagreement, and analyst forecast dispersion. Fourth, we test whether predicted disagreement increases are associated with standard equity risk measures.

4.1. Prediction Performance and Feature Importance

We first evaluate out-of-sample performance in predicting whether ESG rating disagreement will increase over the subsequent fiscal year. Table 2 reports AUC for each ESG dimension across the seven test years. Mean AUC ranges from 0.649 for Overall ESG and Social to 0.667 for Governance, with Environmental at 0.655.¹⁹ All values exceed 0.5, indicating predictive ability beyond random classification. Differences across dimensions are negligible, whereas year-to-year variation is more pronounced, as for instance 2019 shows the strongest prediction across all ESG dimensions.

[Insert Table 2 about here]

Next, we examine which features drive these predictions. Figure 4 presents SAGE values across the four ESG dimensions. The baseline disagreement level is the most informative predictor for all ESG dimensions. Secondary features, however, provide meaningful incremental contributions to classification performance, with the second-ranked feature achieving approximately 40–50% of the top feature's importance.

The composition of these secondary predictors varies across dimensions. For Overall ESG, incident counts across all three pillars rank among the top contributors after baseline disagreement, with Environmental incident changes appearing most frequently, followed by Social and Governance incident changes. External audit, lagged disagreement changes, news coverage, and disclosure changes provide further contributions. Within the Environmental dimension, disclosure changes emerge as the second most important

¹⁹Results are very similar when weighting by the number of firms in each year.

predictor, followed by lagged disagreement changes, Environmental incident changes, and news coverage. For Social, the pattern shifts, with external audit ranking second, followed by lagged disagreement changes, Social incident changes, news coverage, Corporate Social Responsibility (CSR) report issuance, and disclosure changes. Within the Governance dimension, news coverage ranks second, followed by CSR report issuance, Governance incident changes, disclosure changes, lagged disagreement changes, and external audit.

[Insert Figure 4 about here]

Following Zografopoulos et al. (2025), the bottom row of Figure 4 reports feature selection rates, defined as the percentage of test years in which each feature ranked among the top three by SAGE value. Baseline disagreement appears in the top three in 71% of test years for all four ESG dimensions, confirming its consistent importance. By contrast, secondary features differ across dimensions, though within each pillar the top-ranked secondary contributor appears in the top three in 43 to 57% of test years. For Overall ESG, Environmental incident changes appear in the top three in 43% of test years, with Social incident changes, Governance incident changes, external audit, lagged disagreement changes, news coverage, and disclosure changes each appearing in 29%. Within the Environmental dimension, disclosure changes rank among the top three most frequently at 57%, followed by Environmental incident changes and news coverage at 43%. For Social, external audit reaches the highest secondary frequency at 57%, with lagged disagreement changes and Social incident changes each appearing in 43%. For Governance, news coverage, CSR report issuance, lagged disagreement changes, and external audit each appear in the top three in 43% of test years.

While these secondary features vary across dimensions, baseline disagreement remains the top-ranked predictor for all four ESG dimensions. A natural question is what makes it so informative. One candidate is mean reversion. If firms with high disagreement tend to experience subsequent decreases and vice versa, the baseline level would be highly predictive of the direction of change.

To test for this, we estimate²⁰

²⁰The mean reversion regression uses the balanced measure $D_{i,\tau}^{\text{bal}}$, which restricts to provider pairs with coverage in consecutive years, to ensure that estimated dynamics reflect genuine reassessment rather than mechanical effects of rater entry or exit. The ML classifier uses $D_{i,\tau}$ computed from all available pairs as a predictive feature, since at portfolio formation the set of continuing pairs is not yet known.

$$\Delta D_{i,\tau-1} = \beta D_{i,\tau-1}^{\text{bal}} + \mu_{\tau-1} + \phi_{j(i)} + \varepsilon_{i,\tau-1} \quad (13)$$

where $D_{i,\tau-1}^{\text{bal}}$ is the disagreement level for firm i in fiscal year $\tau - 1$, and $\Delta D_{i,\tau-1} = D_{i,\tau}^{\text{bal}} - D_{i,\tau-1}^{\text{bal}}$ is the subsequent change, both computed using only provider pairs with coverage in consecutive years as defined in Equation 4. Year and Fama-French 12 industry fixed effects are denoted $\mu_{\tau-1}$ and $\phi_{j(i)}$, respectively. A negative coefficient $\beta < 0$ indicates mean reversion: higher baseline levels predict subsequent decreases. Rearranging yields $D_{i,\tau}^{\text{bal}} = \rho D_{i,\tau-1}^{\text{bal}} + \mu_{\tau-1} + \phi_{j(i)} + \varepsilon_{i,\tau-1}$ with persistence parameter $\rho = 1 + \beta$. For $0 < \rho < 1$, the implied half-life, $\ln(0.5)/\ln(\rho)$, measures the average time for a shock to decay by half.

[Insert Table 3 about here]

Panel A of Table 3 reports the results for the full 2010–2023 sample. The results confirm that the coefficient on lagged disagreement is negative and highly significant for all four ESG dimensions, consistent with mean-reverting dynamics. The Governance pillar exhibits the strongest mean reversion ($\hat{\beta} = -0.672$, $t = -43.86$), followed by Social ($\hat{\beta} = -0.448$), Overall ESG ($\hat{\beta} = -0.373$), and Environmental ($\hat{\beta} = -0.362$). Consistent with these estimates, all persistence parameters satisfy $0 < \hat{\rho} < 1$, with implied half-lives ranging from 0.6 years for Governance to 1.5 years for Environmental and Overall ESG. Panel B confirms that these dynamics are qualitatively similar during the 2017–2023 out-of-sample period, with coefficients remaining negative and highly significant across all pillars, though mean reversion is somewhat weaker in magnitude (e.g., half-life for the Environmental dimension increases from 1.5 to 2.6 years).

The regression provides evidence that mean reversion exists on average, but does not reveal whether the relationship is uniform across the disagreement distribution. To examine this, we sort firms into quintiles based on $D_{i,\tau-1}^{\text{bal}}$ within each fiscal year and compute the probability of a subsequent increase for each quintile. Figure 5 plots the spread between the extreme quintiles over time:

$$\text{Spread}_{\tau} = P(\Delta D_{i,\tau-1} > 0 \mid Q_5) - P(\Delta D_{i,\tau-1} > 0 \mid Q_1) \quad (14)$$

where Q_1 denotes the lowest disagreement quintile and Q_5 the highest. A negative spread indicates that high-disagreement firms are less likely to experience further increases than low-disagreement firms.

[Insert Figure 5 about here]

The resulting spreads are consistently negative across all pillars and years. In particular, for Overall ESG, firms in the lowest quintile have an average 78% probability of a subsequent increase, compared to 29% for firms in the highest quintile, a spread of -49 percentage points. The Governance dimension exhibits the largest average spread (-60 percentage points), consistent with its stronger mean reversion coefficient, while Environmental (-43 percentage points) and Social (-50 percentage points) fall between these extremes.

To further characterize these dynamics, Figure B.1 and Figure B.2 in the Appendix present transition probabilities between consecutive increases and decreases, confirming that reversals exceed persistence across all pillars and industries.

Taken together, mean-reverting dynamics are consistent with baseline disagreement ranking highest in the SAGE decomposition, with the secondary features providing additional classification performance.

4.2. Predicted Disagreement Increases and Stock Returns

Given the evidence that ESG rating disagreement changes exhibit out-of-sample predictability, we now examine whether the predicted probability of disagreement increase is associated with subsequent stock returns using the panel regressions and portfolio sorts described in Sections 3.2 and 3.3.

4.2.1. Panel Regressions

To assess the incremental contribution of the predicted probability of disagreement increase relative to baseline disagreement levels, Table 4 presents results for the full out-of-sample period. Throughout, all specifications control for size, book-to-market, profitability, momentum, beta, volatility, analyst forecast dispersion, and ESG rating level, with industry and formation-year fixed effects and standard errors clustered by firm.

Column (1) includes only the disagreement level $D_{i,\tau}$. Across ESG dimensions, the resulting coefficients are negative but insignificant for Overall ESG (-0.053), Environmental (-0.097), and Governance (-0.020), with only Social reaching marginal significance (-0.111 , $p < 0.10$).

[Insert Table 4 about here]

To test whether predicted disagreement dynamics carry return-predictive content beyond the baseline level, Column (2) includes $\hat{p}_{i,t}$ instead of $D_{i,\tau}$. The results are markedly different. For instance, the Overall ESG coefficient is 0.096 ($t = 2.80$). In economic terms, because $\hat{p}_{i,t} \in [0, 1]$ and returns are measured in decimal units, a 10 percentage point higher predicted probability is associated with approximately 96 basis points higher annual return. This pattern extends across all ESG dimensions, with coefficients consistently positive and significant, ranging from 0.096 for Environmental ($t = 2.74$) to 0.161 for Social ($t = 3.42$).

Column (3) includes $D_{i,\tau}$ and $\hat{p}_{i,t}$ simultaneously. If $\hat{p}_{i,t}$ contained no incremental information beyond the baseline level $D_{i,\tau}$, we would expect its coefficient to be close to zero once $D_{i,\tau}$ is included. Instead, $\hat{p}_{i,t}$ remains positive and statistically significant for all four ESG dimensions, suggesting that $\hat{p}_{i,t}$ contains return-predictive variation beyond what is captured by the baseline level. Moreover, the behavior of the level coefficient differs across dimensions. For Overall ESG, Environmental, and Social, we do not find evidence that $D_{i,\tau}$ has incremental predictive power once $\hat{p}_{i,t}$ is included. The Governance pillar exhibits a distinct pattern, as both variables enter significantly and the level coefficient reverses sign, from -0.020 ($t = -0.34$) to 0.152 ($t = 2.08$), once $\hat{p}_{i,t}$ is included.

Column (4) removes the within-year linear component of $\hat{p}_{i,t}$ explained by $D_{i,\tau}$. Specifically, we regress $\hat{p}_{i,t}$ on $D_{i,\tau}$ separately within each formation year and extract the residual:

$$\hat{p}_{i,t}^{\perp} = \hat{p}_{i,t} - \hat{\gamma}_{0,t} - \hat{\gamma}_{1,t} D_{i,\tau} \quad (15)$$

Regressing returns on $\hat{p}_{i,t}^{\perp}$ with the full control set and fixed effects, but excluding $D_{i,\tau}$, yields positive and significant coefficients across all ESG dimensions, ranging from 0.145 for Overall ESG ($t = 2.74$) to 0.274 for Environmental ($t = 3.87$). This suggests that $\hat{p}_{i,t}$ is associated with returns beyond the within-year linear component explained by baseline disagreement.²¹

A natural concern is that the return-predictive content of $\hat{p}_{i,t}$ may simply reflect nonlinear functions of the baseline disagreement level. The SAGE decomposition in Section 4.1 identifies $D_{i,\tau}$ as the most informative predictor, and because the MLP maps

²¹Adjusted R^2 exceeds 24% across specifications, which may appear high for a cross-sectional return regression. This explanatory power derives largely from formation-year fixed effects, which absorb common return variation within each annual cohort.

inputs through successive nonlinear transformations, $\hat{p}_{i,t}^\perp$ may retain higher-order functions of $D_{i,\tau}$ even after linear orthogonalization. To examine this, Columns (5) through (7) extend the specification by including $D_{i,\tau}$, $D_{i,\tau}^2$, and $D_{i,\tau}^3$.

The results in Column (5) indicate that no polynomial term reaches significance for any ESG dimension, suggesting that polynomial functions of the disagreement level do not provide incremental return predictability. To test whether these polynomial controls absorb the predictive content of $\hat{p}_{i,t}$, Column (6) adds it to the specification. The predicted probability retains significance for all pillars (t -statistics of 2.77 to 3.36), while no polynomial term reaches significance at the 5% level, though the Environmental linear term is marginally significant at the 10% level. Lastly, Column (7) orthogonalizes $\hat{p}_{i,t}$ to $D_{i,\tau}$, $D_{i,\tau}^2$, and $D_{i,\tau}^3$ within each formation year, removing polynomial content up to cubic order. The resulting residual $\hat{p}_{i,t}^{\perp\perp}$ remains significant across all pillars (t -statistics of 2.51 to 4.41). Since $\hat{p}_{i,t}^{\perp\perp}$ is by construction orthogonal to polynomial functions of $D_{i,\tau}$ up to cubic order, the return-predictive content of $\hat{p}_{i,t}$ does not appear to be driven solely by nonlinear transformations of the baseline level.

Taken together, our results provide evidence that $\hat{p}_{i,t}$ contains return-relevant information beyond what baseline disagreement levels capture, including polynomial transformations of those levels. When disagreement levels enter without $\hat{p}_{i,t}$, coefficients fail to reach significance at the 5% level for any ESG dimension, with only Social marginally significant at 10% and negative in sign. This contrasts with Gibson Brandon et al. (2021), who document a positive disagreement-return relation in S&P 500 firms from 2010 to 2017 using seven raters, with effects driven primarily by the Environmental dimension. Our finding instead aligns with Horky et al. (2024), who study EuroStoxx 600 firms from 2016 to 2022 using three raters, and Anselmi and Petrella (2025), who examine US and European markets from 2010 to 2020 using two raters. Both find that level effects vanish once standard firm characteristics are controlled, and these discrepancies across studies may reflect differences in sample composition, time period, or the number of raters used to construct disagreement measures.

4.2.2. Portfolio Sorts

We next translate the regression evidence into a portfolio setting. At the end of each June, we sort firms into terciles based on $\hat{p}_{i,t}$ and hold the resulting portfolios through June of the following year.

To evaluate the resulting return patterns, Table 5 presents average monthly excess returns, Sharpe ratios, and alphas from Fama-French five-factor plus momentum (FF6) regressions for the full sample period alongside Early (2017 to 2019) and Late (2020 to 2023) subperiods. We focus on the full-sample results here and return to subsample findings in Section 4.2.3. Panel A covers EW portfolios and Panel B covers VW portfolios, with all t -statistics computed using Newey and West (1987) standard errors with 12 lags.

[Insert Table 5 about here]

The results show that the VW H–L spread generates positive FF6 alphas across all four ESG dimensions. The Social and Governance pillars produce the largest spreads at 46 and 47 basis points monthly, respectively, both significant at the 1% level and corresponding to approximately 5.5% annually. The Environmental and Overall ESG spreads are smaller at 27 and 25 basis points, with the former significant at the 1% level and the latter at the 10% level.

To understand what drives these spreads, we examine the individual tercile FF6 alphas. For the Overall ESG sort, the High tercile earns a significant 34 basis points while the Low tercile earns an insignificant 9 basis points, hence the spread derives entirely from the long side. However, the Mid tercile earns –14 basis points, yielding a non-monotonic pattern in which the extremes outperform the middle. The Social dimension presents a different picture, as the spread derives from both legs, with the High tercile earning a significant 24 basis points and the Low tercile earning a significant negative alpha, though the Mid tercile's higher point estimate of 27 basis points does not reach significance. In contrast, the Governance and Environmental dimensions both exhibit monotonic FF6 alphas increasing from Low to Mid to High tercile. The Governance spread is significant at the 1% level, with alphas ranging from –19 to 24 to 27 basis points, while the Environmental spread is similarly significant though neither individual tercile alpha reaches significance on its own.

EW portfolios yield significant spreads for all four ESG dimensions, ranging from 23 to 35 basis points monthly, with stronger monotonicity than the VW results. FF6 alphas increase from Low to Mid to High for Overall ESG, Environmental, and Governance, though Social is the exception, where the Mid tercile exceeds the High tercile. Across all four dimensions, the source of the EW spreads shifts toward the Low tercile, where firms with low predicted probability of disagreement increase earn significant negative FF6 alphas for Overall ESG, Environmental, and Social, with Governance reaching significance

at the 10% level. By contrast, High tercile EW alphas are generally insignificant. This difference from the VW results, where spreads are driven by the High tercile for the Overall ESG and Governance portfolio sorts, is consistent with size-related heterogeneity in the return association.

Comparing with prior research, Gibson Brandon et al. (2021) obtain equal-weighted factor model alphas of 23 to 27 basis points monthly from quintile sorts on industry-adjusted ESG disagreement levels, with Environmental disagreement generating similar alphas (21 to 25 basis points) while Social and Governance long-short strategies produce no significant returns. Similarly, Avramov et al. (2022) obtain a monthly CAPM alpha of 23 basis points from sorting on Overall ESG rating uncertainty levels. By contrast, Anselmi and Petrella (2025) construct portfolios sorted by realized past disagreement changes and report mixed patterns across pillars without consistent return predictability, whereas our sorts on predicted disagreement dynamics generate significant H–L spreads.

Taken together, the portfolio evidence corroborates the directional panel regression findings. The predicted probability of disagreement increase is positively associated with subsequent returns across all four ESG dimensions in the full out-of-sample period, with effects that persist after controlling for standard risk factors.

4.2.3. Robustness

To assess whether these patterns are stable over time, we split the sample into an Early period (2017 to 2019) and a Late period (2020 to 2023). Table 6 presents the full panel regression specification with all controls across both subperiods, and the corresponding portfolio sort results appear in the Early and Late columns of Table 5.

[Insert Table 6 about here]

The results show that temporal stability varies across dimensions. For the Environmental dimension, regression coefficients remain significant in both Early and Late, and VW portfolio spreads increase from 33 to 55 basis points. By contrast, EW spreads are insignificant in the Early period before reaching significance in the Late period. This strengthening may reflect the growing salience of climate risk to institutional investors (Krueger et al., 2020) and aligns with evidence that carbon-related return premia increased following the Paris Agreement (Bolton and Kacperczyk, 2021; Agliardi et al., 2023). The Social dimension exhibits a similar strengthening, as VW alphas also increase across

subperiods (46 to 57 basis points, both significant), and EW alphas remain significant throughout. Moreover, regression coefficients remain significant at the 10% level in the Late period.

For the Governance pillar, regression coefficients and VW portfolio spreads instead decline and lose statistical significance in the Late period, while EW spreads, though significant in the full out-of-sample period, do not reach significance in either subperiod individually. Overall ESG exhibits a broadly similar pattern in regressions and VW sorts, though EW spreads remain significant throughout. This attenuation likely reflects compositional effects, as the aggregate measure combines pillars with opposing temporal dynamics, and the strengthening Environmental effect and the weakening Governance effect partially offset one another. That said, the subperiod analysis necessarily reduces statistical power given that the full out-of-sample period spans only seven formation years.

Further, to assess implementability, we compute net-of-cost returns using Corwin and Schultz (2012) bid-ask spread estimates. Specifically, we deduct one-way trading costs for each entering and exiting position at annual rebalancing. Table 7 presents the results for the full out-of-sample period. The EW H–L alphas remain significant for the Overall ESG, Environmental, and Social dimensions (29, 32, and 19 basis points, respectively). Similarly, VW H–L alphas remain significant for the Environmental, Social, and Governance dimensions (24, 43, and 43 basis points, respectively). Across both weighting schemes, the modest reduction relative to gross-of-cost spreads is consistent with the low turnover implied by annual rebalancing.

[Insert Table 7 about here]

Finally, to assess sensitivity to model choice, we estimate a logistic model with identical features, preprocessing, and walk-forward design. Table B.1 presents classification performance. The logistic model achieves mean AUC of 0.593 to 0.612 across dimensions, compared to 0.649 to 0.667 for the MLP (Table 2). However, the logistic model's predicted probabilities do not reach significance in the full-sample panel regression for any dimension (Table B.2), and VW H-L FF6 alphas are uniformly insignificant (Table B.3). That said, EW spreads reach significance for the Environmental and Social dimensions, though with smaller magnitudes than the MLP-based sorts. As both models receive identical inputs, the return-predictive content appears to reside in nonlinear feature combinations that a

linear specification cannot capture.²²

Taken together, the predicted probability of disagreement increase from the MLP is positively associated with subsequent returns across panel regressions and portfolio sorts in the full out-of-sample period, with the Environmental and Social dimensions providing the most stable evidence across subperiods and weighting schemes. Baseline disagreement levels generally do not reach significance when standard controls are included. Moreover, predicted probabilities from the logistic model do not produce consistent return associations, suggesting that the signal requires nonlinear modeling capacity.

These findings are also consistent with an economic interpretation. Gibson Brandon et al. (2021) suggest that ESG rating disagreement reflects Knightian uncertainty about ESG information, consistent with evidence that investors exhibit ambiguity when processing ESG ratings from different providers (Bachner, 2025). If disagreement captures this form of uncertainty, then ambiguity-averse investors may require higher expected returns when they anticipate greater uncertainty about future information quality (Epstein and Schneider, 2008), and our results are consistent with a setting in which such anticipation would carry pricing implications beyond those captured by baseline disagreement levels. That said, we view these results as consistent with an ambiguity-based interpretation rather than as a direct test of the mechanism.

4.3. Does the Return Association Vary with Firm Characteristics?

The preceding analysis provides evidence for a positive association between $\hat{p}_{i,t}$ and subsequent returns in the full out-of-sample period. We now examine whether this association varies with observable firm characteristics by interacting $\hat{p}_{i,t}$ with four conditioning variables, namely firm size, baseline disagreement level, pillar-specific ESG rating level, and analyst forecast dispersion.

The choice of moderators is motivated by two broad considerations. The first relates to firm visibility, as larger firms attract more attention from ESG-sensitive investors, analysts, and media (Starks et al., 2017; Krueger et al., 2020), so if these actors drive the return association, it should be stronger among more visible firms. The second relates to whether the firm already faces heightened ESG scrutiny or fundamental uncertainty, either of which

²²Consistent with this interpretation, untabulated results using gradient boosted trees (XGBoost) and random forests, estimated with identical walk-forward cross-validation, produce classification performance comparable to the MLP and generate significant return associations of similar magnitude.

could compound with anticipated disagreement increases. Firms with elevated baseline disagreement may experience compounding uncertainty when the predicted probability of further increase is also high. Similarly, firms with weak ESG performance face greater scrutiny on sustainability issues (Hartzmark and Sussman, 2019) that could sharpen the market's response. Finally, high analyst forecast dispersion signals fundamental cash-flow uncertainty (Zhang, 2006) that may interact with ESG-specific uncertainty to amplify the association.

To test these predictions, Table 8 presents the interaction estimates. We demean each conditioning variable within formation year so that the coefficient on $\hat{p}_{i,t}$ represents the association evaluated at the cross-sectional mean of each moderator. Columns (1) through (4) enter each interaction individually, and Column (5) enters all four simultaneously.

The size interaction is positive and significant for the Overall ESG (0.100, $t = 2.51$), Social (0.162, $t = 3.78$), and Governance dimensions (0.080, $t = 2.92$), with all three surviving the joint specification. By contrast, the Environmental dimension is the exception, where the size interaction is insignificant. The positive size interaction for three of four dimensions is consistent with the return association being larger among firms that receive more attention from ESG-sensitive market participants.

A similar pattern holds for analyst forecast dispersion. The interaction coefficient is positive and significant for the Social (0.094, $t = 4.16$), Overall ESG (0.053, $t = 1.96$), and Governance dimensions (0.025, $t = 2.10$), with all three surviving the joint specification, consistent with fundamental and ESG-specific uncertainty compounding.

The Environmental dimension is the only one for which the ESG rating level moderates the return association. The interaction is negative and significant (-0.413 , $t = -2.33$), suggesting that the return association with the predicted probability of Environmental disagreement increase is stronger among firms with low Environmental ratings. This result survives the joint specification (-0.396 , $t = -2.12$). For the remaining three dimensions, the ESG rating level interaction is insignificant at the 5% level.

Lastly, baseline disagreement does not moderate the return association for any dimension, whether entered individually or in the joint specification.

[Insert Table 8 about here]

4.4. Are Predicted Disagreement Increases Associated with Equity Risk?

For completeness, and in the spirit of Gibson Brandon et al. (2021), we examine whether predicted disagreement increases are associated with standard equity risk measures. If the return association documented above reflected compensation for conventional risk exposure, we would expect firms with higher $\hat{p}_{i,t}$ to also exhibit higher volatility or beta during the holding period. We estimate

$$\text{Risk}_{i,t \rightarrow t+1} = \alpha + \delta_1 \hat{p}_{i,t} + \boldsymbol{\gamma}^\top \mathbf{Z}_{i,t} + \phi_{j(i)} + \mu_t + \varepsilon_{i,t}, \quad (16)$$

where $\text{Risk}_{i,t \rightarrow t+1}$ is one of three considered risk measures, namely total volatility, idiosyncratic volatility, and market beta. $\mathbf{Z}_{i,t}$ collects the set of firm-level controls, $\phi_{j(i)}$ and μ_t denote industry and formation-year fixed effects, and standard errors are clustered at the firm level.

Table 9 presents the results. Across all three risk measures and all four ESG dimensions, δ_1 is insignificant at conventional levels, with no t -statistic exceeding 1.48 in absolute value. This mirrors the finding in Gibson Brandon et al. (2021) that ESG rating disagreement levels are largely unrelated to standard equity risk measures, and is consistent with the view that ESG rating disagreement captures a form of information uncertainty distinct from conventional risk exposure.

[Insert Table 9 about here]

5. Conclusion

This paper examines whether the dynamics of ESG rating disagreement contain return-relevant information beyond what baseline disagreement levels capture. Using ESG ratings from four major providers for S&P 500 firms over the 2010 to 2023 period, we train neural network classifiers with walk-forward cross-validation and examine the association between the resulting out-of-sample predicted probabilities and subsequent stock returns.

We provide evidence that the predicted probability of disagreement increase is positively and significantly associated with subsequent returns in the full out-of-sample period for all four ESG dimensions in panel regressions and under most portfolio sort specifications, while baseline disagreement levels generally are not. This association remains significant after orthogonalization with respect to polynomial functions of the disagreement level.

Furthermore, a logistic model estimated with identical features does not produce consistent return associations, suggesting that the signal depends on nonlinear modeling capacity. The predicted probability is also not associated with total volatility, idiosyncratic volatility, or market beta for any dimension, suggesting that the return association does not reflect conventional risk compensation.

The return association is also moderated by firm characteristics. It is significantly stronger among larger firms and those with higher analyst forecast dispersion for three of four dimensions, and for the Environmental dimension it is stronger among firms with weaker Environmental ratings. The baseline disagreement level, by contrast, does not moderate the association for any dimension. Across subperiods, the Environmental and Social dimensions provide the most stable evidence and also retain significant net-of-cost portfolio spreads under both equal- and value-weighting.

SAGE-based decomposition attributes the classifier's predictive performance to mean-reverting disagreement dynamics captured by the baseline level and to dimension-specific secondary features such as disclosure practices, ESG incidents, news coverage, and lagged disagreement changes.

Overall, these findings are consistent with an ambiguity-based interpretation. If ESG rating disagreement captures the form of Knightian uncertainty argued by Gibson Brandon et al. (2021), then the predicted probability of disagreement increase may reflect greater anticipated uncertainty about the quality of signals investors receive, and under ambiguity aversion such uncertainty would affect expected returns (Epstein and Schneider, 2008). That said, our analysis documents the return association rather than providing a direct test of this mechanism. Moreover, our sample is limited to large-cap U.S. firms and an out-of-sample period that spans seven formation years. Whether similar patterns hold in other markets, for different rater combinations, or over longer horizons remains for future research.

6. Tables

Table 1: ESG RATING PROVIDER COVERAGE

Year	Ref.		MSCI		Sust.		S&P		Total	MR Cov.			Pairs
	N	%	N	%	N	%	N	%		≥2	≥3	4	
2010	402	96	388	93	0	0	0	0	417	375	0	0	0.9
2011	393	96	392	96	0	0	0	0	410	377	0	0	0.9
2012	390	95	401	97	0	0	0	0	412	383	0	0	0.9
2013	388	94	408	99	0	0	0	0	413	388	0	0	0.9
2014	389	94	408	98	0	0	0	0	415	389	0	0	0.9
2015	385	94	402	98	0	0	0	0	409	385	0	0	0.9
2016	381	94	396	98	0	0	0	0	405	380	0	0	0.9
2017	379	94	394	98	0	0	355	88	404	384	350	0	2.7
2018	377	94	392	98	0	0	364	91	401	383	358	0	2.7
2019	374	94	389	97	0	0	368	92	400	381	360	0	2.8
2020	378	94	392	97	366	91	373	92	404	383	374	358	5.5
2021	382	94	396	97	376	92	378	93	407	389	382	363	5.5
2022	378	94	391	98	376	94	382	95	401	385	380	369	5.6
2023	380	94	395	98	380	94	384	95	403	388	382	372	5.6

Notes: Number of S&P 500 firms (N) and coverage rate (%) by ESG rating provider. Year denotes portfolio formation year (last trading day of June); coverage reflects fiscal year $\tau = t - 1$ ESG ratings available at formation. Ref. = Refinitiv (LSEG), Sust. = Sustainalytics, S&P = S&P Global. Total denotes the number of S&P 500 constituent firms covered by at least one ESG rating provider in a given year. MR Cov. = multi-rater coverage: firms rated by at least 2, at least 3, or all 4 providers. Pairs = average number of provider pairs per firm across all index constituents (maximum 6).

Table 2: OUT-OF-SAMPLE CLASSIFICATION PERFORMANCE BY ESG DIMENSION

Training Period	Test Year	Overall ESG	Environmental	Social	Governance
2010–2016	2017	0.667	0.658	0.648	0.673
2010–2017	2018	0.636	0.643	0.647	0.681
2010–2018	2019	0.754	0.747	0.736	0.768
2010–2019	2020	0.613	0.618	0.623	0.641
2010–2020	2021	0.634	0.641	0.637	0.644
2010–2021	2022	0.618	0.627	0.636	0.608
2010–2022	2023	0.621	0.653	0.614	0.657
Mean		0.649	0.655	0.649	0.667

Notes: Out-of-sample AUC for the MLP classifier predicting whether firm-level ESG rating disagreement increases over the subsequent fiscal year. The model uses walk-forward cross-validation with expanding training windows beginning in 2010. Architecture and hyperparameters are selected as described in Section 3.1.2. AUC = 0.5 indicates no discriminative ability beyond random classification; AUC = 1.0 indicates perfect separation.

Table 3: MEAN REVERSION IN ESG RATING DISAGREEMENT

	Overall ESG	Environmental	Social	Governance
Panel A: Full Sample (2010–2023)				
$D_{i,\tau-1}^{\text{bal}}$	-0.373*** (-22.17)	-0.362*** (-24.92)	-0.448*** (-23.68)	-0.672*** (-43.86)
$\hat{\rho}$	0.627	0.638	0.552	0.328
Half-life (years)	1.5	1.5	1.2	0.6
N	5,126	5,126	5,126	5,126
Adj. R^2 (%)	21.4	20.7	24.9	35.6
Panel B: Out-of-Sample Period (2017–2023)				
$D_{i,\tau-1}^{\text{bal}}$	-0.301*** (-20.22)	-0.230*** (-17.55)	-0.300*** (-20.25)	-0.455*** (-25.83)
$\hat{\rho}$	0.699	0.770	0.700	0.545
Half-life (years)	1.9	2.6	1.9	1.1
N	2,585	2,585	2,585	2,585
Adj. R^2 (%)	17.0	12.8	16.7	24.8
Ind. + Year FE	✓			

Notes: This table tests for mean reversion in ESG rating disagreement. Panel A uses the full 2010–2023 sample; Panel B restricts to the 2017–2023 out-of-sample period. Each panel reports coefficients from: $\Delta D_{i,\tau-1} = \beta D_{i,\tau-1}^{\text{bal}} + \mu_{\tau-1} + \phi_{j(i)} + \varepsilon_{i,\tau-1}$, where $D_{i,\tau-1}^{\text{bal}}$ is the disagreement level for firm i in fiscal year $\tau - 1$, and $\Delta D_{i,\tau-1} = D_{i,\tau}^{\text{bal}} - D_{i,\tau-1}^{\text{bal}}$ is the subsequent change, both computed using only provider pairs with coverage in consecutive years. A negative β indicates mean reversion. The persistence parameter $\hat{\rho} = 1 + \hat{\beta}$ is obtained by rearranging to the level specification $D_{i,\tau}^{\text{bal}} = \rho D_{i,\tau-1}^{\text{bal}} + \mu_{\tau-1} + \phi_{j(i)} + \varepsilon_{i,\tau-1}$. For $0 < \hat{\rho} < 1$, the half-life $\ln(0.5)/\ln(\hat{\rho})$ measures the time in years for a shock to decay by half. All estimates satisfy $0 < \hat{\rho} < 1$. Year and Fama-French 12 industry fixed effects included throughout. Standard errors clustered by firm. t -statistics in parentheses. Bold indicates $p < 0.05$. ***, **, * denote 1%, 5%, 10%.

Table 4: RETURN PREDICTABILITY: LEVEL VS. PREDICTED CHANGE

Panel A: Overall ESG and Environmental														
	Overall ESG							Environmental						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$D_{i,\tau}$	-0.053 (-0.71)		0.119 (1.12)		-0.516 (-1.08)	0.050 (0.10)		-0.097 (-1.52)		0.059 (0.58)		0.650 (1.15)	1.221* (1.95)	
$D^2_{i,\tau}$					2.303 (1.00)	0.966 (0.40)						-2.512 (-0.89)	-3.902 (-1.32)	
$D^3_{i,\tau}$					-3.126 (-0.95)	-2.004 (-0.59)						1.996 (0.47)	3.457 (0.78)	
$\hat{p}_{i,t}$		0.096*** (2.80)	0.139*** (2.94)			0.151*** (3.09)			0.096*** (2.74)	0.121** (2.15)			0.163*** (2.78)	
$\hat{p}^\perp_{i,t}$				0.145*** (2.74)							0.274*** (3.87)			
$\hat{p}^{\perp\perp}_{i,t}$							0.195*** (3.31)							0.343*** (4.41)
N							2,278							
Adj. R^2 (%)	24.26	24.46	24.49	24.49	24.26	24.50	24.60	24.36	24.48	24.47	24.75	24.42	24.62	24.90
Controls							✓							
Ind. + Year FE							✓							
Panel B: Social and Governance														
	Social							Governance						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$D_{i,\tau}$	-0.111* (-1.78)		0.037 (0.43)		0.399 (0.57)	0.481 (0.68)		-0.020 (-0.34)		0.152** (2.08)		0.200 (0.41)	0.287 (0.59)	
$D^2_{i,\tau}$					-2.170 (-0.62)	-1.644 (-0.47)						-0.914 (-0.44)	-0.212 (-0.10)	
$D^3_{i,\tau}$					2.607 (0.51)	1.666 (0.32)						1.087 (0.41)	-0.207 (-0.08)	
$\hat{p}_{i,t}$		0.161*** (3.42)	0.181*** (2.76)			0.183*** (2.77)			0.126*** (2.86)	0.180*** (3.29)			0.184*** (3.36)	
$\hat{p}^\perp_{i,t}$				0.251*** (2.94)							0.190*** (3.36)			
$\hat{p}^{\perp\perp}_{i,t}$							0.221** (2.51)							0.232*** (3.86)
N							2,278							
Adj. R^2 (%)	24.35	24.56	24.54	24.53	24.32	24.51	24.45	24.25	24.57	24.66	24.57	24.19	24.62	24.69
Controls							✓							
Ind. + Year FE							✓							

Notes: This table tests whether the predicted probability of disagreement increase captures return-relevant information beyond nonlinear functions of the baseline disagreement level. The dependent variable is the buy-and-hold return from July of year t through June of year $t + 1$. Column (1) includes only the baseline disagreement level $D_{i,\tau}$. Column (2) includes only the predicted probability $\hat{p}_{i,t}$. Column (3) includes both $D_{i,\tau}$ and $\hat{p}_{i,t}$. Column (4) uses $\hat{p}_{i,t}^\perp$, the residual from regressing $\hat{p}_{i,t}$ on $D_{i,\tau}$ within each formation year, which removes the within-year linear association between the predicted probability and the baseline level. Column (5) includes $D_{i,\tau}$, $D_{i,\tau}^2$, and $D_{i,\tau}^3$ to test whether polynomial functions of the disagreement level predict returns. Column (6) adds $\hat{p}_{i,t}$ to the polynomial specification in Column (5). Column (7) uses $\hat{p}_{i,t}^{\perp\perp}$, the residual from regressing $\hat{p}_{i,t}$ on $D_{i,\tau}$, $D_{i,\tau}^2$, and $D_{i,\tau}^3$ within each formation year, which removes polynomial content up to the third order. All specifications control for size, log book-to-market, gross profitability, momentum, analyst forecast dispersion, beta, volatility, and ESG rating level, and include Fama-French 12 industry and formation-year fixed effects. Standard errors are clustered by firm. t -statistics are reported in parentheses. Bold indicates $p < 0.05$. ***, **, * denote significance at 1%, 5%, and 10%.

Table 5: PORTFOLIO PERFORMANCE: ALL TERCILES WITH RETURNS AND ALPHAS

	Overall ESG			Environmental			Social			Governance		
	Full	Early	Late	Full	Early	Late	Full	Early	Late	Full	Early	Late
Panel A: Equal-Weighted Portfolios												
<i>Low Portfolio</i>												
Mean Return	0.81*	0.50	1.04	0.73*	0.51	0.90	0.73*	0.50	0.91	0.79**	0.50	1.02*
	(1.74)	(1.25)	(1.42)	(1.89)	(1.25)	(1.51)	(1.77)	(1.34)	(1.38)	(2.22)	(1.31)	(1.88)
Sharpe Ratio	0.14	0.08	0.19	0.14	0.09	0.18	0.14	0.09	0.18	0.15	0.09	0.21
FF6 α	-0.23***	-0.18***	-0.33***	-0.29**	-0.12	-0.42**	-0.29***	-0.20**	-0.41***	-0.20*	-0.15	-0.26
	(-2.80)	(-3.32)	(-3.21)	(-2.04)	(-1.12)	(-2.24)	(-3.06)	(-2.07)	(-3.49)	(-1.70)	(-1.61)	(-1.59)
<i>Mid Portfolio</i>												
Mean Return	0.85**	0.65	1.01	0.98**	0.72*	1.18*	1.03**	0.78**	1.22*	0.93**	0.60	1.17*
	(2.10)	(1.63)	(1.55)	(2.23)	(1.88)	(1.68)	(2.56)	(2.13)	(1.91)	(2.32)	(1.46)	(1.84)
Sharpe Ratio	0.16	0.12	0.20	0.18	0.12	0.23	0.20	0.14	0.24	0.17	0.10	0.22
FF6 α	-0.16	-0.07	-0.32	-0.07	-0.11	-0.14	0.05	0.03	-0.09	-0.14	-0.07	-0.20
	(-1.25)	(-0.66)	(-1.60)	(-0.87)	(-1.20)	(-1.36)	(0.39)	(0.51)	(-0.44)	(-1.03)	(-0.58)	(-1.04)
<i>High Portfolio</i>												
Mean Return	1.14***	0.79**	1.41***	1.09***	0.70*	1.38**	1.04**	0.65	1.33**	1.08**	0.84**	1.27*
	(3.13)	(2.02)	(2.63)	(2.68)	(1.83)	(2.20)	(2.51)	(1.50)	(2.16)	(2.29)	(2.07)	(1.72)
Sharpe Ratio	0.21	0.13	0.27	0.20	0.13	0.25	0.18	0.11	0.25	0.19	0.14	0.23
FF6 α	0.10	0.04	0.04	0.07	0.03	-0.05	-0.05	-0.04	-0.11	0.05	0.01	-0.15
	(0.76)	(0.48)	(0.23)	(0.53)	(0.20)	(-0.23)	(-0.47)	(-0.48)	(-0.74)	(0.54)	(0.18)	(-1.33)
<i>High-Low Spread</i>												
Mean Return	0.34**	0.29***	0.37	0.35***	0.19**	0.48***	0.30***	0.15	0.42***	0.29*	0.34**	0.25
	(2.21)	(2.65)	(1.48)	(2.92)	(2.01)	(2.64)	(3.25)	(1.60)	(4.02)	(1.83)	(2.47)	(1.04)
Sharpe Ratio	0.25	0.24	0.26	0.26	0.15	0.34	0.29	0.14	0.43	0.23	0.30	0.18
FF6 α	0.33**	0.22***	0.37***	0.35***	0.14	0.37***	0.23***	0.16**	0.30***	0.25**	0.16	0.11
	(2.47)	(2.88)	(2.88)	(3.47)	(0.81)	(3.44)	(2.81)	(2.33)	(4.47)	(2.23)	(1.61)	(1.18)
Panel B: Value-Weighted Portfolios												
<i>Low Portfolio</i>												
Mean Return	1.21***	1.14***	1.27*	0.91**	0.80**	0.98	0.94**	0.80**	1.04*	0.97***	0.71**	1.22**
	(2.72)	(2.84)	(1.84)	(2.05)	(2.05)	(1.36)	(2.40)	(2.11)	(1.76)	(2.75)	(1.98)	(2.09)
Sharpe Ratio	0.23	0.22	0.23	0.17	0.15	0.18	0.19	0.16	0.20	0.20	0.14	0.25
FF6 α	0.09	0.03	0.03	-0.04	-0.09	-0.10	-0.23***	-0.34***	-0.18	-0.19*	-0.14	-0.11
	(1.20)	(0.41)	(0.24)	(-0.32)	(-0.57)	(-0.63)	(-2.84)	(-6.08)	(-1.48)	(-1.89)	(-1.43)	(-0.77)
<i>Mid Portfolio</i>												
Mean Return	0.83**	0.67	0.98	1.22***	1.31***	1.14*	1.26***	1.39***	1.28*	1.34***	1.18***	1.55**
	(2.10)	(1.64)	(1.46)	(2.77)	(3.17)	(1.65)	(2.87)	(3.86)	(1.65)	(3.72)	(3.48)	(2.43)
Sharpe Ratio	0.17	0.14	0.19	0.24	0.24	0.23	0.25	0.31	0.23	0.28	0.26	0.30
FF6 α	-0.14	-0.21	-0.03	0.11	0.15	-0.02	0.27	0.51***	0.21	0.24*	0.34**	0.36**
	(-0.84)	(-0.99)	(-0.15)	(0.88)	(0.88)	(-0.16)	(1.38)	(3.40)	(0.60)	(1.78)	(2.24)	(2.33)
<i>High Portfolio</i>												
Mean Return	1.53***	1.43***	1.64**	1.43***	1.09***	1.73***	1.40***	1.00**	1.72**	1.35**	1.39***	1.22
	(3.69)	(3.49)	(2.47)	(3.94)	(3.08)	(2.85)	(3.06)	(2.23)	(2.38)	(2.49)	(3.08)	(1.50)
Sharpe Ratio	0.29	0.28	0.30	0.28	0.24	0.31	0.25	0.18	0.31	0.24	0.26	0.21
FF6 α	0.34***	0.47***	0.38***	0.23	0.24	0.44**	0.24**	0.12	0.39***	0.27**	0.14	0.11
	(2.94)	(4.06)	(2.88)	(1.51)	(1.40)	(2.56)	(2.03)	(1.12)	(3.65)	(2.13)	(1.17)	(0.62)
<i>High-Low Spread</i>												
Mean Return	0.32	0.28	0.37	0.52***	0.29*	0.76***	0.46*	0.20	0.68**	0.38	0.68***	0.01
	(1.52)	(0.77)	(1.62)	(3.28)	(1.84)	(3.33)	(1.72)	(0.52)	(2.10)	(1.26)	(2.65)	(0.01)
Sharpe Ratio	0.15	0.15	0.16	0.24	0.16	0.29	0.21	0.10	0.27	0.15	0.38	0.00
FF6 α	0.25*	0.44**	0.35*	0.27***	0.33**	0.55***	0.46***	0.46***	0.57***	0.47***	0.28**	0.22
	(1.74)	(2.51)	(1.93)	(2.73)	(2.15)	(4.34)	(3.21)	(3.58)	(3.05)	(2.75)	(1.96)	(0.85)

Notes: Complete portfolio performance across all terciles, pillars, and periods. Each year, firms are sorted into terciles based on the predicted probability of a dimension-specific disagreement increase for four ESG dimensions (Overall ESG, Environmental, Social, Governance) across three periods (Full 2017–2023, Early 2017–2019, Late 2020–2023). For each portfolio (Low, Mid, High, H–L), we report: mean monthly excess return (%), Sharpe ratio, and Fama-French (FF) 6-factor alpha. Panel A shows equal-weighted portfolios; Panel B shows value-weighted portfolios. Returns and alphas in % per month. Newey and West (1987) *t*-statistics (12 lags) in parentheses. Bold indicates significance at 5%. ***, **, * denote 1%, 5%, 10% significance.

Table 6: RETURN PREDICTABILITY: FULL SPECIFICATION

	Overall ESG			Environmental			Social			Governance		
	F	E	L	F	E	L	F	E	L	F	E	L
Panel A: Main Variable of Interest												
$\hat{p}_{i,t}$	0.139*** (2.94)	0.111** (2.05)	0.166 (1.35)	0.121** (2.15)	0.167** (2.16)	0.198** (2.27)	0.181*** (2.76)	0.223** (2.08)	0.196* (1.91)	0.180*** (3.29)	0.175** (2.56)	0.117 (1.62)
Panel B: Firm Characteristics												
Size	-0.000 (-0.06)	0.012 (1.27)	-0.004 (-0.38)	0.003 (0.35)	0.014 (1.45)	-0.001 (-0.12)	-0.000 (-0.01)	0.015* (1.66)	-0.005 (-0.49)	0.001 (0.14)	0.017** (1.99)	-0.005 (-0.47)
B/M	-0.013** (-1.99)	-0.031*** (-3.14)	-0.003 (-0.34)	-0.012** (-2.00)	-0.029*** (-3.01)	-0.005 (-0.45)	-0.012** (-1.98)	-0.029*** (-3.01)	-0.004 (-0.39)	-0.012** (-1.99)	-0.030*** (-3.06)	-0.004 (-0.40)
Prof	0.065 (1.56)	0.120** (2.44)	-0.020 (-0.29)	0.056 (1.33)	0.111** (2.28)	-0.026 (-0.38)	0.047 (1.15)	0.108** (2.18)	-0.035 (-0.52)	0.048 (1.15)	0.106** (2.18)	-0.032 (-0.47)
Mom	-0.073*** (-3.52)	-0.006 (-0.21)	-0.145*** (-5.18)	-0.068*** (-3.28)	-0.000 (-0.01)	-0.139*** (-4.96)	-0.069*** (-3.35)	0.001 (0.05)	-0.142*** (-5.07)	-0.070*** (-3.40)	-0.005 (-0.19)	-0.141*** (-4.99)
Disp	0.007*** (2.91)	-0.080 (-1.31)	0.007*** (3.02)	0.007*** (2.73)	-0.080 (-1.28)	0.006*** (2.59)	0.007*** (3.08)	-0.080 (-1.36)	0.007*** (3.12)	0.005** (2.31)	-0.087 (-1.48)	0.005*** (2.67)
Beta	0.011 (1.15)	-0.003 (-0.47)	0.040 (1.39)	0.011 (1.17)	-0.003 (-0.47)	0.038 (1.33)	0.012 (1.24)	-0.002 (-0.32)	0.043 (1.49)	0.011 (1.18)	-0.002 (-0.24)	0.040 (1.38)
Vol	0.004 (0.36)	-0.008 (-0.58)	0.015 (0.93)	0.004 (0.39)	-0.008 (-0.60)	0.013 (0.84)	0.005 (0.50)	-0.005 (-0.36)	0.016 (1.00)	0.005 (0.50)	-0.005 (-0.38)	0.015 (0.94)
Panel C: ESG Variables												
D Level	0.119 (1.12)	0.156 (1.22)	0.019 (0.12)	0.059 (0.58)	0.323** (2.02)	-0.054 (-0.42)	0.037 (0.43)	0.012 (0.13)	0.102 (0.60)	0.152** (2.08)	0.146* (1.76)	0.068 (0.52)
ESG Level	-0.011 (-0.37)	0.014 (0.37)	-0.039 (-0.82)	-0.043 (-1.34)	0.002 (0.05)	-0.092 (-1.63)	0.019 (0.67)	0.035 (0.90)	-0.017 (-0.34)	0.004 (0.11)	-0.004 (-0.08)	0.010 (0.19)
Model Statistics												
N	2,278	981	1,297	2,278	981	1,297	2,278	981	1,297	2,278	981	1,297
Adj. R^2 (%)	24.490	12.348	32.043	24.466	12.298	32.438	24.539	12.313	32.119	24.660	12.451	32.018
Ind + Year FE												✓

Notes: This table reports panel regressions of annual stock returns on predicted probability of an ESG disagreement increase and controls. The dependent variable is the buy-and-hold return from July of year t through June of year $t + 1$. $\hat{p}_{i,t}$ is the predicted probability of pillar-specific disagreement increase from our machine learning model, estimated using only information available at portfolio formation. F denotes the full out-of-sample period (2017–2023); E denotes the early period (2017–2019); L denotes the late period (2020–2023). Size is the natural log of market capitalization. B/M is the natural log of the book-to-market ratio. Prof is revenues minus cost of goods sold, scaled by total assets. Mom is the cumulative return over the 11 months ending in May of formation year t , skipping the month immediately prior to formation (June of year t). Disp is the standard deviation of analyst EPS forecasts scaled by the absolute value of mean forecast. Beta and Vol are estimated over 12 months prior to portfolio formation. All pillar-specific specifications use corresponding pillar measures: D Level denotes the baseline disagreement level for that pillar, and ESG Level denotes the average rating level for that pillar. All specifications include industry and formation-year fixed effects. t -statistics based on standard errors clustered at the firm level are reported in parentheses. Bold indicates significance at the 5% level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 7: PORTFOLIO PERFORMANCE NET OF TRANSACTION COSTS

	Overall ESG	Environmental	Social	Governance
Panel A: Equal-Weighted Portfolios				
<i>Low Portfolio</i>				
Mean Return	0.78*	0.71*	0.71*	0.77**
	(1.69)	(1.84)	(1.72)	(2.16)
Sharpe Ratio	0.14	0.13	0.13	0.15
FF6 α	-0.25***	-0.30**	-0.31***	-0.22*
	(-3.07)	(-2.16)	(-3.31)	(-1.89)
<i>Mid Portfolio</i>				
Mean Return	0.82**	0.96**	1.00**	0.90**
	(2.04)	(2.18)	(2.50)	(2.26)
Sharpe Ratio	0.16	0.18	0.19	0.17
FF6 α	-0.19	-0.09	0.02	-0.16
	(-1.45)	(-1.14)	(0.19)	(-1.21)
<i>High Portfolio</i>				
Mean Return	1.12***	1.07***	1.01**	1.06**
	(3.07)	(2.63)	(2.45)	(2.24)
Sharpe Ratio	0.21	0.19	0.18	0.19
FF6 α	0.08	0.05	-0.07	0.03
	(0.61)	(0.38)	(-0.68)	(0.29)
<i>High-Low Spread</i>				
Mean Return	0.29*	0.31***	0.25***	0.24
	(1.90)	(2.64)	(2.70)	(1.53)
Sharpe Ratio	0.21	0.23	0.25	0.19
FF6 α	0.29**	0.32***	0.19**	0.21*
	(2.23)	(3.20)	(2.40)	(1.78)
Panel B: Value-Weighted Portfolios				
<i>Low Portfolio</i>				
Mean Return	1.19***	0.89**	0.92**	0.95***
	(2.68)	(2.00)	(2.36)	(2.69)
Sharpe Ratio	0.23	0.16	0.18	0.19
FF6 α	0.07	-0.06	-0.24***	-0.21**
	(0.96)	(-0.47)	(-3.09)	(-2.10)
<i>Mid Portfolio</i>				
Mean Return	0.81**	1.20***	1.23***	1.32***
	(2.03)	(2.72)	(2.81)	(3.65)
Sharpe Ratio	0.16	0.23	0.25	0.27
FF6 α	-0.16	0.09	0.24	0.21
	(-0.98)	(0.71)	(1.26)	(1.62)
<i>High Portfolio</i>				
Mean Return	1.50***	1.41***	1.38***	1.31**
	(3.63)	(3.89)	(3.01)	(2.45)
Sharpe Ratio	0.29	0.28	0.25	0.23
FF6 α	0.32***	0.21	0.22*	0.25**
	(2.79)	(1.42)	(1.87)	(1.99)
<i>High-Low Spread</i>				
Mean Return	0.27	0.48***	0.42	0.32
	(1.29)	(3.04)	(1.55)	(1.11)
Sharpe Ratio	0.13	0.22	0.19	0.13
FF6 α	0.22	0.24**	0.43***	0.43**
	(1.47)	(2.44)	(2.94)	(2.51)

Notes: This table reports net-of-transaction-cost portfolio performance across all terciles and pillars for the full out-of-sample period (2017–2023). At each annual rebalancing, we deduct one-way transaction costs based on Corwin and Schultz (2012) bid-ask spread estimates for each entering and exiting position. Portfolios are formed each June by sorting firms into terciles on the predicted probability of a dimension-specific disagreement increase across four ESG dimensions (Overall ESG, Environmental, Social, Governance). For each portfolio (Low, Mid, High, H–L), we report mean monthly excess return (%), Sharpe ratio, and Fama–French 6-factor alpha. Panel A shows equal-weighted portfolios; Panel B shows value-weighted portfolios. Returns and alphas in % per month. Newey and West (1987) t -statistics (12 lags) in parentheses. Bold indicates significance at the 5% level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 8: INTERACTION OF PREDICTED DISAGREEMENT INCREASE WITH FIRM CHARACTERISTICS

	Size (1)	Disagreement (2)	ESG Level (3)	Analyst Disp. (4)	All (5)
<i>Panel A: Overall ESG</i>					
$\hat{p}_{i,t}$	0.143*** (3.02)	0.127*** (2.60)	0.143*** (3.04)	0.139*** (2.95)	0.134*** (2.75)
$\hat{p}_{i,t} \times \text{Size}$	0.100** (2.51)				0.089** (2.17)
$\hat{p}_{i,t} \times \text{Disagr.}$		-0.358 (-0.87)			-0.345 (-0.86)
$\hat{p}_{i,t} \times \text{ESG Level}$			0.263* (1.72)		0.152 (0.96)
$\hat{p}_{i,t} \times \text{Analyst Disp.}$				0.053** (1.96)	0.058** (2.23)
Adj. R^2 (%)	24.72	24.50	24.55	24.51	24.74
<i>Panel B: Environmental</i>					
$\hat{p}_{i,t}$	0.115** (2.06)	0.128** (2.18)	0.119** (2.13)	0.121** (2.15)	0.125** (2.15)
$\hat{p}_{i,t} \times \text{Size}$	-0.051 (-1.42)				-0.023 (-0.62)
$\hat{p}_{i,t} \times \text{Disagr.}$		0.208 (0.58)			0.260 (0.74)
$\hat{p}_{i,t} \times \text{ESG Level}$			-0.413** (-2.33)		-0.396** (-2.12)
$\hat{p}_{i,t} \times \text{Analyst Disp.}$				-0.038 (-0.64)	-0.052 (-0.85)
Adj. R^2 (%)	24.50	24.45	24.58	24.45	24.54
<i>Panel C: Social</i>					
$\hat{p}_{i,t}$	0.169*** (2.63)	0.186*** (2.84)	0.180*** (2.75)	0.181*** (2.80)	0.170*** (2.66)
$\hat{p}_{i,t} \times \text{Size}$	0.162*** (3.78)				0.171*** (3.88)
$\hat{p}_{i,t} \times \text{Disagr.}$		0.713 (1.41)			0.646 (1.30)
$\hat{p}_{i,t} \times \text{ESG Level}$			-0.063 (-0.26)		-0.287 (-1.14)
$\hat{p}_{i,t} \times \text{Analyst Disp.}$				0.094*** (4.16)	0.090*** (3.46)
Adj. R^2 (%)	24.91	24.59	24.51	24.65	25.05
<i>Panel D: Governance</i>					
$\hat{p}_{i,t}$	0.187*** (3.45)	0.173*** (3.13)	0.180*** (3.29)	0.180*** (3.28)	0.180*** (3.29)
$\hat{p}_{i,t} \times \text{Size}$	0.080*** (2.92)				0.081*** (2.93)
$\hat{p}_{i,t} \times \text{Disagr.}$		0.276 (1.04)			0.280 (1.08)
$\hat{p}_{i,t} \times \text{ESG Level}$			-0.036 (-0.23)		-0.087 (-0.56)
$\hat{p}_{i,t} \times \text{Analyst Disp.}$				0.025** (2.10)	0.026** (2.03)
Adj. R^2 (%)	24.92	24.66	24.63	24.67	24.91
Controls			✓		
Industry + Year FE			✓		
N			2,278		

Notes: This table reports regressions of annual buy-and-hold returns on the predicted probability of disagreement increase ($\hat{p}_{i,t}$) and its interaction with firm characteristics. Each panel uses the pillar-specific predicted probability, disagreement level, and ESG rating level corresponding to that dimension. Conditioning variables are demeaned within formation year before interaction, so the coefficient on $\hat{p}$ represents the return association evaluated at the cross-sectional mean of each conditioning variable. Columns (1)–(4) enter each interaction individually; Column (5) enters all four simultaneously. Controls include log market capitalization, log book-to-market, gross profitability, momentum, market beta, total volatility, analyst forecast dispersion, pillar-specific disagreement level, and pillar-specific ESG rating level. Standard errors are clustered by firm; t -statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

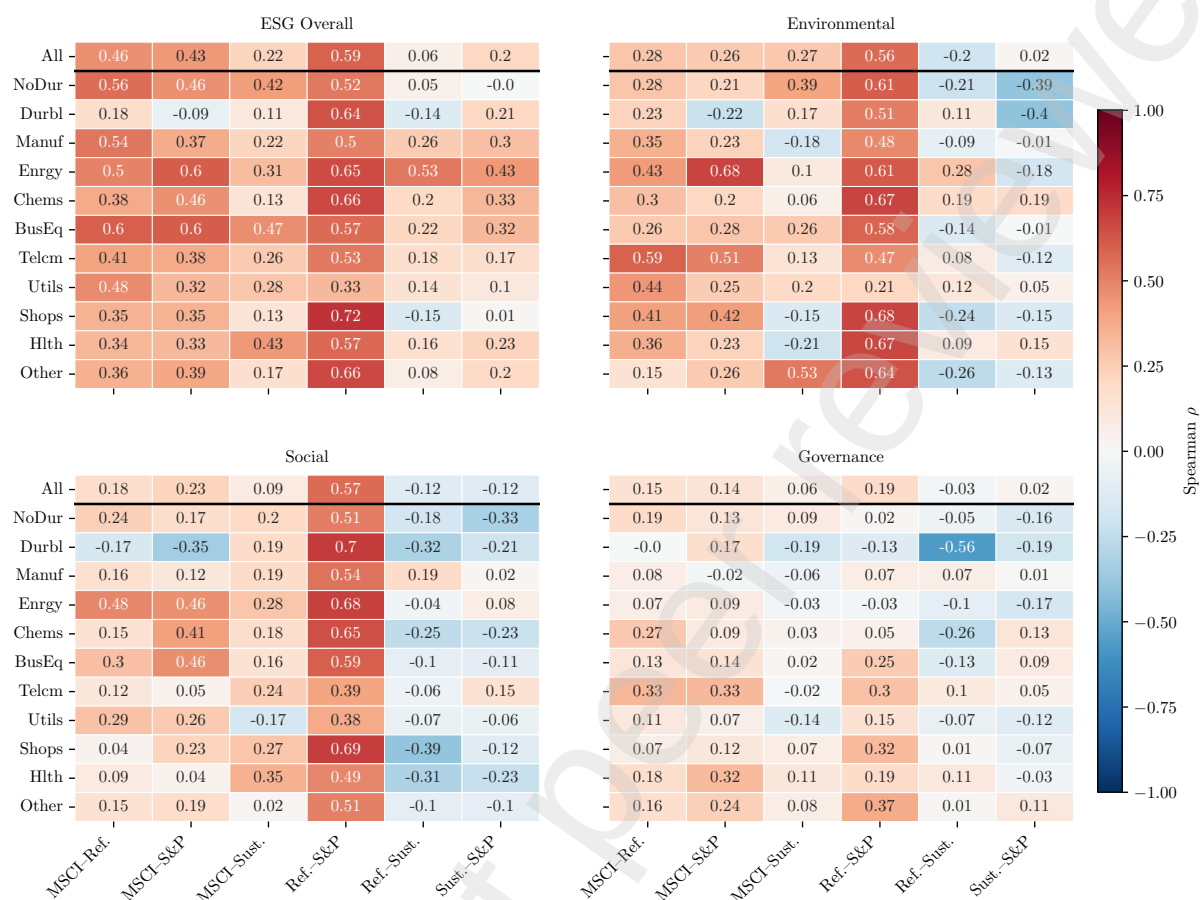
Table 9: PREDICTED DISAGREEMENT CHANGE AND HOLDING-PERIOD EQUITY RISK

	Total Volatility	Idiosyncratic Volatility	Beta
<i>Panel A: Overall ESG</i>			
$\hat{p}_{i,t}$	0.009 (0.40)	−0.000 (−0.01)	0.031 (0.30)
<i>Panel B: Environmental</i>			
$\hat{p}_{i,t}$	0.004 (0.25)	−0.001 (−0.10)	0.077 (0.79)
<i>Panel C: Social</i>			
$\hat{p}_{i,t}$	−0.025 (−1.27)	−0.025 (−1.48)	−0.003 (−0.02)
<i>Panel D: Governance</i>			
$\hat{p}_{i,t}$	0.008 (0.56)	0.011 (1.05)	0.058 (0.67)
Controls		✓	
Ind. + Year FE		✓	
N		2,278	

Notes: Each cell reports $\hat{\delta}_1$ from Equation 16. Dependent variables are computed from monthly returns over the twelve-month holding period (July of year t through June of year $t + 1$). Total volatility is the annualized standard deviation of monthly returns. Idiosyncratic volatility is the annualized standard deviation of residuals from a Fama and French (1993) three-factor regression. Beta is the loading on the excess market return. Building on the specification of Gibson Brandon et al. (2021), controls include log market capitalization, log book-to-market, gross profitability, momentum (cumulative return over the 11 months ending in May of year t , skipping the month immediately prior to formation), analyst forecast dispersion, leverage, number of business segments, turnover, and institutional ownership (see Table A.1 for definitions). All specifications include Fama–French 12 industry and formation-year fixed effects. t -statistics based on firm-clustered standard errors in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

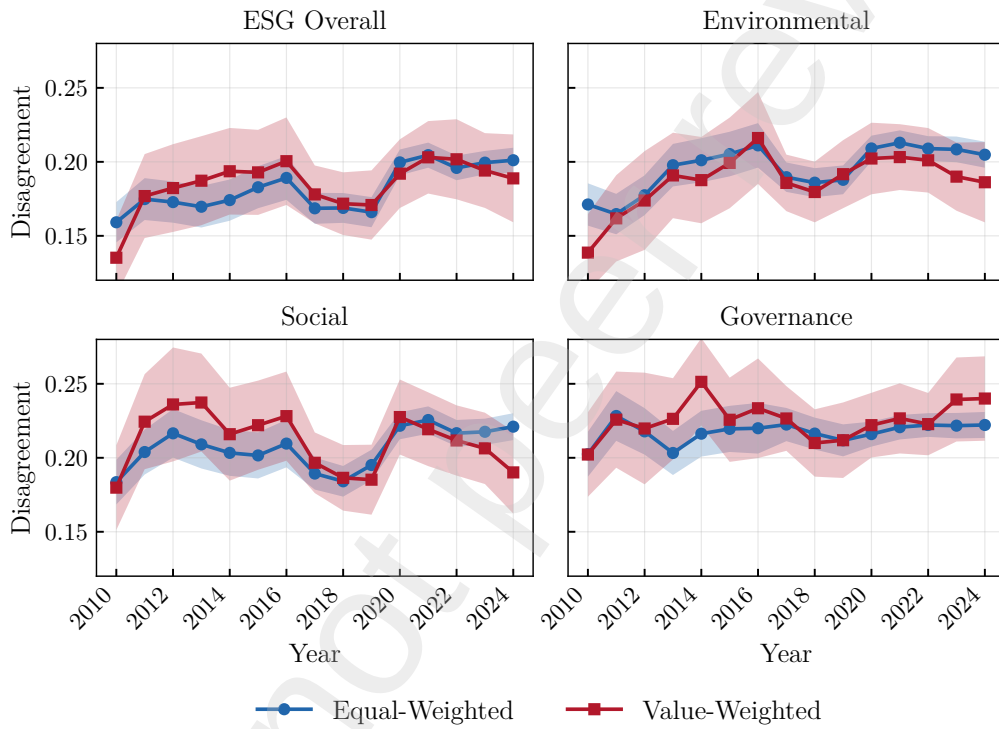
7. Figures

Figure 1: ESG RATING CORRELATIONS BY INDUSTRY



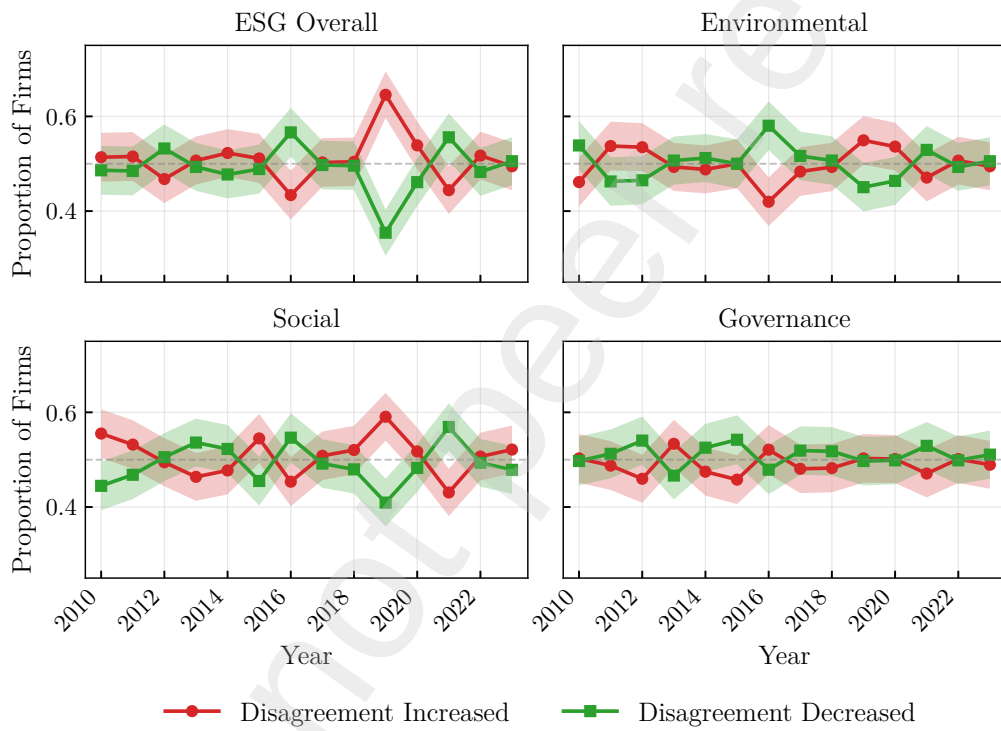
Notes: Spearman rank correlations between ESG ratings from MSCI, Refinitiv/LSEG (Ref.), Sustainalytics (Sust.), and S&P Global (S&P) by Fama-French 12 industry. Each panel reports pairwise correlations for a different pillar. Rows correspond to industries, with "All" denoting the pooled sample. Sustainalytics scores are multiplied by -1 to align with other providers where higher values indicate better ESG performance.

Figure 2: TIME SERIES OF ESG RATING DISAGREEMENT



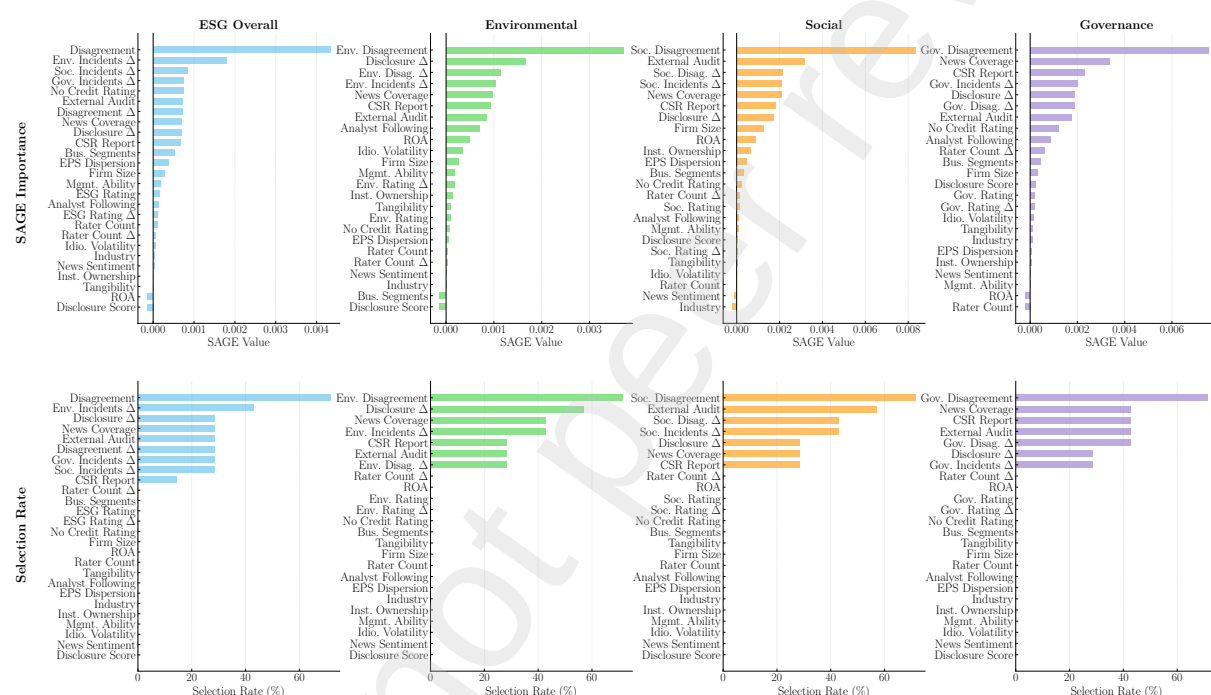
Notes: Each panel plots annual cross-sectional mean disagreement for Overall ESG and the Environmental, Social, and Governance pillars separately. We collect ESG ratings from four providers: MSCI, Refinitiv (LSEG), Sustainalytics, and S&P Global. For each provider pair (p, q) in each fiscal year, we identify firms rated by both providers and compute percentile ranks within this common coverage set. Pairwise disagreement for firm i is $d_{i,\tau}^{pq} = |\pi_{i,\tau}^{p|pq} - \pi_{i,\tau}^{q|pq}|/\sqrt{2}$, where $\pi_{i,\tau}^{p|pq}$ is the percentile rank from provider p within the (p, q) overlap; firm-level disagreement averages across all available provider pairs following Avramov et al. (2022). The equal-weighted series (blue) computes the simple cross-sectional mean of firm-level disagreement each year. The value-weighted series (red) weights each firm by its market capitalization as of June 30. Shaded regions represent 95% confidence intervals.

Figure 3: DIRECTION OF ESG RATING DISAGREEMENT CHANGES



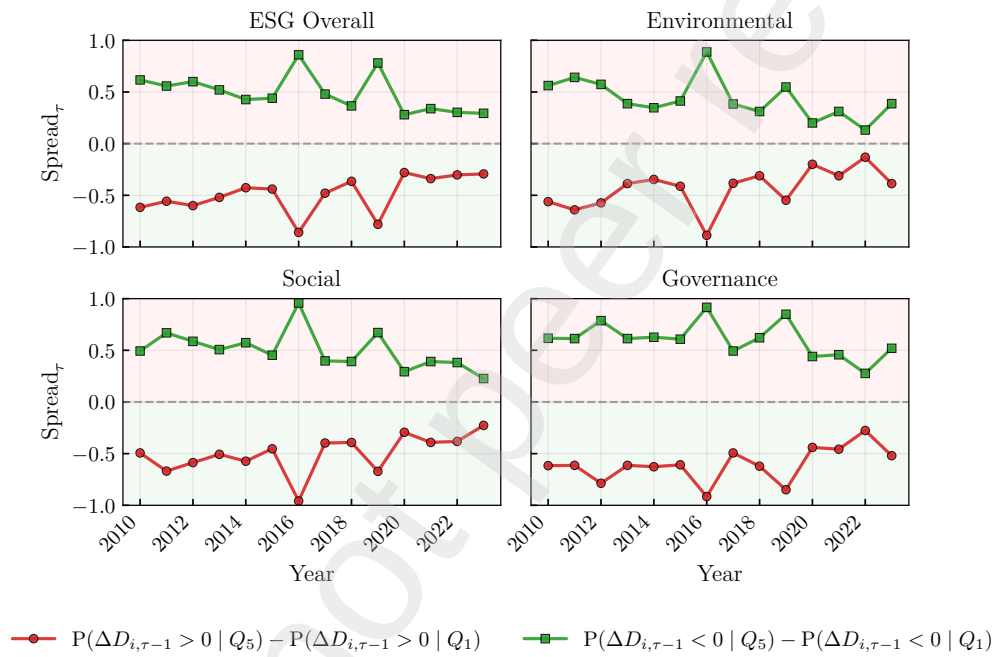
Notes: Each panel plots the annual proportion of firms experiencing disagreement increases (red) versus decreases (green) for Overall ESG and the Environmental, Social, and Governance pillars separately. We collect ESG ratings from four providers: MSCI, Refinitiv (LSEG), Sustainalytics, and S&P Global. For each firm-year, we compute the change in disagreement as $\Delta D_{i,\tau} = D_{i,\tau+1}^{\text{bal}} - D_{i,\tau}^{\text{bal}}$, using only provider pairs with coverage in both years, where $D_{i,\tau}^{\text{bal}}$ is the average absolute pairwise difference in percentile ranks, normalized by $\sqrt{2}$ following Avramov et al. (2022).

Figure 4: SAGE FEATURE IMPORTANCE AND SELECTION RATES FOR ESG RATING DISAGREEMENT PREDICTION



Notes: This figure presents SAGE (Shapley Additive Global importance) values and feature selection rates from MLP models predicting changes in ESG rating disagreement direction. The top row displays mean SAGE importance values with 95% confidence intervals computed across seven out-of-sample test years (2017–2023) for Overall ESG and the Environmental, Social, and Governance pillars separately. SAGE values quantify each feature's average contribution to reducing cross-entropy loss, representing the expected decrease in prediction error when the feature is observed rather than marginalized out. Blue bars indicate positive contributions; red bars indicate negative contributions. The bottom row shows feature selection rates, defined as the percentage of test years in which each feature ranked among the top three by SAGE value.

Figure 5: MEAN REVERSION IN ESG RATING DISAGREEMENT: LEVEL PREDICTS CHANGE



Notes: Each panel plots the annual probability spread between extreme disagreement quintiles for Overall ESG and the Environmental, Social, and Governance pillars separately. Within each fiscal year, we sort firms into quintiles based on baseline disagreement level $D_{i,t-1}^{bal}$. The red line plots $P(\Delta D_{i,t-1} > 0 | Q_5) - P(\Delta D_{i,t-1} > 0 | Q_1)$, the difference in the probability of a subsequent disagreement increase between the highest and lowest quintiles. The green line plots the corresponding difference for decreases. A negative red spread indicates mean reversion: firms with high baseline disagreement are less likely to experience further increases than firms with low baseline disagreement. The horizontal dashed line at zero indicates no difference across quintiles.

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APPENDIX TO:
ESG RATING DISAGREEMENT DYNAMICS AND STOCK
RETURNS

Appendix A. Additional Descriptives

Table A.1: VARIABLE DEFINITIONS

Variable	Definition
Panel A: ESG Disagreement Variables	
Disagreement Level (Overall) ^{O,R}	Mean pairwise disagreement across all available ESG rater pairs for Overall ESG. Following Avramov et al. (2022), disagreement for each pair is measured using mid-ECDF ranks within the overlap sample. Measured at year $t - 1$. Source: MSCI, Refinitiv (LSEG), Sustainalytics, S&P Global.
Disagreement Change (Overall) ^O	Change in Overall ESG disagreement from year $t - 2$ to year $t - 1$.
Disagreement Level (Env.) ^{E,R}	Mean pairwise disagreement for Environmental pillar at year $t - 1$.
Disagreement Change (Env.) ^E	Change in Environmental disagreement from year $t - 2$ to year $t - 1$.
Disagreement Level (Soc.) ^{S,R}	Mean pairwise disagreement for Social pillar at year $t - 1$.
Disagreement Change (Soc.) ^S	Change in Social disagreement from year $t - 2$ to year $t - 1$.
Disagreement Level (Gov.) ^{G,R}	Mean pairwise disagreement for Governance pillar at year $t - 1$.
Disagreement Change (Gov.) ^G	Change in Governance disagreement from year $t - 2$ to year $t - 1$.
Rating Level (Overall) ^{O,R}	Following Avramov et al. (2022), mean ESG rating across raters, expressed as average mid-ECDF rank. Measured at year $t - 1$.

Continued on next page

Table A.1 (continued)

Variable	Definition
Rating Change (Overall) ^O	Change in average Overall ESG rating from year $t - 2$ to year $t - 1$.
Rating Level (Env.) ^{E,R}	Mean Environmental rating across raters at year $t - 1$.
Rating Change (Env.) ^E	Change in average Environmental rating from year $t - 2$ to year $t - 1$.
Rating Level (Soc.) ^{S,R}	Mean Social rating across raters at year $t - 1$.
Rating Change (Soc.) ^S	Change in average Social rating from year $t - 2$ to year $t - 1$.
Rating Level (Gov.) ^{G,R}	Mean Governance rating across raters at year $t - 1$.
Rating Change (Gov.) ^G	Change in average Governance rating from year $t - 2$ to year $t - 1$.
Number of Raters ^P	Count of ESG raters with coverage of Overall ESG Score (maximum 4) at year $t - 1$.
Number of Raters Change ^P	Change in rater coverage from year $t - 2$ to year $t - 1$ of Overall ESG Score.
Panel B: Target Variables	
Disagreement Increase (Overall) [†]	Binary indicator equal to 1 if Overall ESG disagreement increased from year $t - 1$ to year t .
Disagreement Increase (Env.) [†]	Binary indicator equal to 1 if Environmental disagreement increased from year $t - 1$ to year t .
Disagreement Increase (Soc.) [†]	Binary indicator equal to 1 if Social disagreement increased from year $t - 1$ to year t .
Disagreement Increase (Gov.) [†]	Binary indicator equal to 1 if Governance disagreement increased from year $t - 1$ to year t .

Continued on next page

Table A.1 (continued)

Variable	Definition
Panel C: ESG Disclosure and Incidents	
ESG Disclosure Score ^P	Extent of firm's ESG disclosure practices, ranging from 0.1 to 100, tailored by industry sector. Source: Bloomberg.
Disclosure Score Change ^P	Change in Bloomberg ESG Disclosure Score from year $t - 2$ to year $t - 1$.
CSR Sustainability Report ^{P,†}	Binary indicator equal to 1 if firm published a CSR/sustainability report; set to zero if missing. Source: Refinitiv (LSEG).
CSR External Audit ^{P,†}	Binary indicator equal to 1 if firm's CSR report is subject to independent third-party assurance; set to zero if missing. Source: Refinitiv (LSEG).
Δ RepRisk Env. Count ^{O,E}	Following Li and Wu (2020), change in number of Environmental incidents between the 12-month windows ending June $t - 1$ and June t . Source: RepRisk.
Δ RepRisk Soc. Count ^{O,S}	Following Li and Wu (2020), change in number of Social incidents between the 12-month windows ending June $t - 1$ and June t . Source: RepRisk.
Δ RepRisk Gov. Count ^{O,G}	Following Li and Wu (2020), change in number of Governance incidents between the 12-month windows ending June $t - 1$ and June t . Source: RepRisk.

Continued on next page

Table A.1 (continued)

Variable	Definition
Panel D: Firm Characteristics	
Industry Classification ^P	Fama-French 12 industry classification based on SIC codes. Used as categorical feature in machine learning models; also used to construct industry fixed effects in regression analyses. Source: Compustat and Kenneth French's data library.
Firm Size ^{P,R}	Following Alves et al. (2025), computed as the natural logarithm of the absolute value of the product of end-of-month stock price (CRSP field ALTPRC) and number of shares outstanding (CRSP field SHROUT) in millions of USD. Source: CRSP.
Book-to-Market ^R	Following Alves et al. (2025), natural logarithm of the ratio of book value of equity to market capitalization. Source: Compustat.
Number of Business Segments ^{P,R}	Count of unique business segments, excluding geographic segments. Source: Compustat Segments.
Asset Tangibility ^P	Following Alves et al. (2025), property, plant, and equipment divided by total assets. Source: Compustat.
Return on Assets ^P	Net income divided by total assets. Source: Compustat.
Gross Profitability ^R	Following Alves et al. (2025), computed as (Revenue – COGS) / Total assets. Source: Compustat.

Continued on next page

Table A.1 (continued)

Variable	Definition
Leverage ^R	Following Alves et al. (2025), ratio of total debt (Compustat items DLTT + DLC) to total assets (AT). Source: Compustat.
Panel E: Return-Based Measures	
12-Month Momentum ^R	Following Jegadeesh and Titman (1993), cumulative return over the 11 months ending in May of formation year t , skipping the month immediately prior to formation (June of year t) to avoid short-term reversal effects. Source: CRSP.
Market Beta ^R	Loading on market factor from Fama and French (1993) 3-factor model estimated over the 12 months prior to portfolio formation in June of year t . Source: CRSP/Kenneth French's data library.
Total Volatility ^R	Following Gibson Brandon et al. (2021), annualized standard deviation of daily returns estimated over the 12 months prior to portfolio formation in June of year t . Source: CRSP.
Idiosyncratic Volatility ^P	Annualized standard deviation of residuals from Fama and French (1993) 3-factor regression over the 12 months prior to portfolio formation in June of year t . Source: CRSP/Kenneth French's data library.

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Table A.1 (continued)

Variable	Definition
Turnover ^R	Following Kimbrough et al. (2024), annual average of the natural logarithm of daily share turnover, defined as daily trading volume scaled by number of shares outstanding, over the 12 months prior to portfolio formation. Source: CRSP.
Panel F: Analyst, Institutional, and Other	
Analyst Following ^P	Following Christensen et al. (2022), natural logarithm of (1 + number of analysts providing EPS forecasts). Source: I/B/E/S.
Analyst Forecast Dispersion ^{P,R}	Standard deviation of analyst EPS forecasts divided by absolute mean forecast. Following Diether et al. (2002), only computed when mean EPS > \$0.10. Source: I/B/E/S.
Institutional Ownership ^{P,R}	Percentage of shares held by institutions (13F filings). Following Lewellen (2011), capped at 100%; set to zero if missing per Christensen et al. (2022). Source: Refinitiv (LSEG).
Managerial Ability Score ^P	Managerial ability measure from Demerjian et al. (2012).
News Coverage ^P	Following Jia et al. (2023), natural logarithm of (1 + news article count) over 12 months preceding formation. Source: RavenPack.
News Sentiment ^P	Following Jia et al. (2023), average composite sentiment score rescaled to [-1, +1]. Source: RavenPack.

Continued on next page

Table A.1 (continued)

Variable	Definition
No Credit Rating ^{P,†}	Following Gibson Brandon et al. (2021), binary indicator equal to 1 if no long-term credit rating is available. Source: Capital IQ.

Notes: ^PPrediction feature used in all pillar models. ^OOverall ESG model only. ^EEnvironmental pillar model only. ^SSocial pillar model only. ^GGovernance pillar model only. ^RControl variable in one or more regression specifications. [†]Binary indicator. Industry classification is additionally used to construct fixed effects in all regression specifications.

Table A.2: SPEARMAN CORRELATION MATRIX

[illegible]

Notes: This table reports spearman correlation coefficients (upper triangle). Significance levels: 1%, 5%, 10%. Variable definitions: (1) Disagreement Level; (2) Disagreement Level (Env.); (3) Disagreement Level (Soc.); (4) Disagreement Level (Gov.); (5) Disagreement Change; (6) Disagreement Change (Env.); (7) Disagreement Change (Soc.); (8) Disagreement Change (Gov.); (9) Rating Level; (10) Rating Level (Env.); (11) Rating Level (Soc.); (12) Rating Level (Gov.); (13) Rating Change; (14) Rating Change (Env.); (15) Rating Change (Soc.); (16) Rating Change (Gov.); (17) Bloomberg Disclosure; (18) Bloomberg Disclosure Change; (19) CSR Report; (20) CSR External Audit; (21) Number of Raters; (22) Number of Raters Change; (23) RepRisk Env. Count; (24) RepRisk Soc. Count; (25) RepRisk Gov. Count; (26) Firm Size; (27) Book-to-Market; (28) Return on Assets; (29) Gross Profitability; (30) Asset Tangibility; (31) Business Segments; (32) Momentum; (33) Market Beta; (34) Total Volatility; (35) Idiosyncratic Volatility; (36) Analyst Following; (37) EPS Dispersion; (38) Institutional Ownership; (39) Managerial Ability; (40) News Coverage; (41) News Sentiment; (42) No Credit Rating; (43) Target (Overall); (44) Target (Env.); (45) Target (Soc.); (46) Target (Gov.).

Table A.3: SUMMARY STATISTICS

Variable	N	Mean	SD	Min	Max	Skew	Kurt
Panel A: ESG Disagreement Variables							
Disagreement Level (Overall)	5,370	0.18	0.12	0.000	0.67	0.84	0.72
Disagreement Change (Overall)	5,136	0.00	0.10	−0.571	0.58	0.10	2.54
Disagreement Level (Env.)	5,370	0.20	0.12	0.000	0.67	0.61	0.12
Disagreement Change (Env.)	5,136	0.00	0.10	−0.571	0.59	0.20	4.05
Disagreement Level (Soc.)	5,370	0.21	0.13	0.000	0.70	0.75	0.33
Disagreement Change (Soc.)	5,136	0.00	0.12	−0.631	0.57	−0.09	2.39
Disagreement Level (Gov.)	5,370	0.22	0.13	0.000	0.67	0.55	0.02
Disagreement Change (Gov.)	5,136	−0.00	0.15	−0.664	0.65	0.07	1.36
Rating Level (Overall)	5,370	0.50	0.23	0.011	1.00	0.03	−0.88
Rating Change (Overall)	5,136	0.00	0.09	−0.512	0.44	0.15	1.84
Rating Level (Env.)	5,370	0.49	0.22	0.018	0.99	−0.02	−0.78
Rating Change (Env.)	5,136	0.00	0.10	−0.671	0.51	0.22	3.31
Rating Level (Soc.)	5,370	0.50	0.21	0.016	0.99	−0.02	−0.73
Rating Change (Soc.)	5,136	0.00	0.12	−0.480	0.58	0.15	2.04
Rating Level (Gov.)	5,370	0.50	0.20	0.006	1.00	−0.09	−0.45
Rating Change (Gov.)	5,136	0.00	0.15	−0.593	0.64	0.01	1.07
Number of Raters	5,370	2.75	0.85	2.00	4.00	0.49	−1.45
Number of Raters Change	5,149	0.14	0.35	0.000	2.00	2.13	2.73
Panel B: Target Variables							
Disagreement Increase (Overall)	5,149	0.51	0.50	0.000	1.00	−0.03	−2.00
Disagreement Increase (Env.)	5,149	0.50	0.50	0.000	1.00	0.01	−2.00
Disagreement Increase (Soc.)	5,149	0.51	0.50	0.000	1.00	−0.03	−2.00
Disagreement Increase (Gov.)	5,149	0.49	0.50	0.000	1.00	0.04	−2.00
Panel C: ESG Disclosure and Incidents							
ESG Disclosure Score	5,296	47.64	13.34	9.18	85.81	0.15	−0.75

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Table A.3 (continued)

Variable	N	Mean	SD	Min	Max	Skew	Kurt
ESG Disclosure Score Change	5,265	1.93	4.18	-41.86	28.32	0.74	7.85
CSR Sustainability Report	5,701	0.62	0.49	0.000	1.00	-0.50	-1.75
CSR External Audit	5,701	0.24	0.43	0.000	1.00	1.22	-0.51
Δ RepRisk Env. Count	4,774	0.16	4.04	-41.00	72.00	1.56	49.11
Δ RepRisk Soc. Count	4,774	0.37	5.68	-42.00	107.0	3.11	53.48
Δ RepRisk Gov. Count	4,774	0.31	3.58	-37.00	47.00	1.84	27.91
Panel D: Firm Characteristics							
Firm Size	5,653	9.96	1.08	7.03	14.74	0.69	0.61
Book-to-Market	5,439	-1.20	0.90	-4.61	1.19	-0.86	1.28
Number of Business Segments	4,566	3.56	2.30	1.00	16.00	0.76	0.59
Asset Tangibility	5,643	0.29	0.25	0.001	0.95	0.93	-0.39
Return on Assets	5,653	0.07	0.08	-1.23	0.51	-1.57	24.51
Gross Profitability	5,653	0.31	0.20	-1.14	1.30	0.99	2.69
Leverage	5,653	0.31	0.20	0.000	2.00	1.96	11.77
Panel E: Return-Based Measures							
12-Month Momentum	5,687	0.17	0.34	-0.905	3.73	1.51	8.05
Market Beta	5,687	1.00	0.32	-0.207	2.47	0.12	0.49
Total Volatility	5,687	0.30	0.12	0.101	1.19	1.63	4.75
Idiosyncratic Volatility	5,687	0.22	0.09	0.081	0.96	1.70	5.24
Turnover	5,687	-4.86	0.51	-6.283	-2.23	0.75	1.22
Panel F: Analyst, Institutional, and Other							
Analyst Following	5,445	2.88	0.44	0.000	4.03	-1.12	3.50
Analyst Forecast Dispersion	5,438	0.28	1.92	0.000	57.65	21.42	505.84
Institutional Ownership (%)	5,701	75.57	31.05	0.000	100.0	-1.74	1.66
Managerial Ability Score	5,015	0.06	0.19	-0.310	0.64	0.76	0.04
News Coverage	4,596	6.69	0.91	0.693	10.65	0.60	1.81

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Table A.3 (continued)

Variable	<i>N</i>	Mean	SD	Min	Max	Skew	Kurt
News Sentiment	4,596	0.01	0.03	−0.164	0.14	−0.78	2.44
No Credit Rating	5,701	0.28	0.45	0.000	1.00	0.97	−1.07

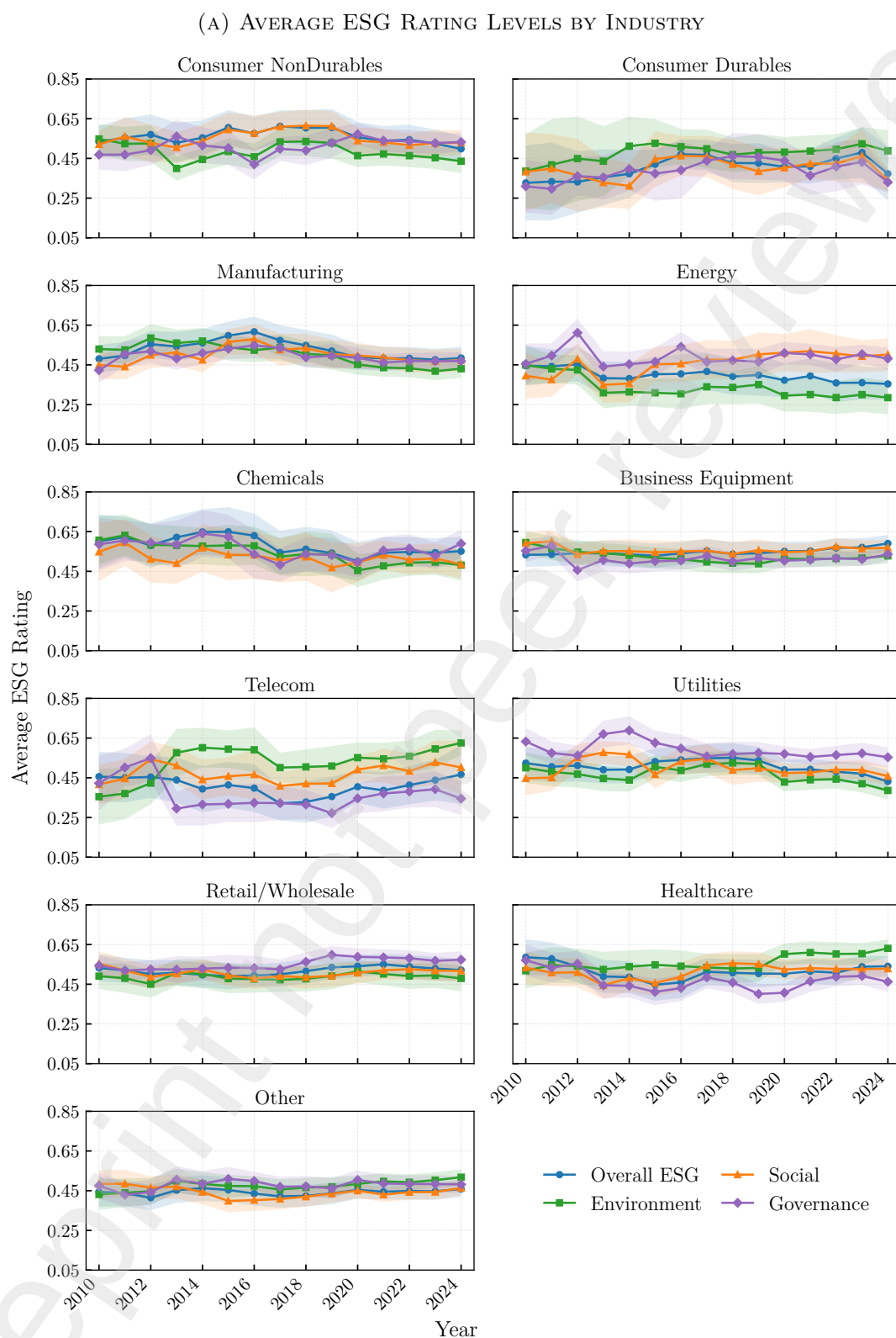
Notes: This table reports summary statistics for all variables used in the analysis. The sample covers S&P 500 firms with coverage from at least one ESG rater ($N = 5,701$) from formation years 2010 to 2023. Env. = Environmental; Soc. = Social; Gov. = Governance. Variable definitions, data sources, and model usage (prediction features vs. regression controls) are detailed in Table A.1.

Table A.4: MLP HYPERPARAMETER SEARCH SPACE

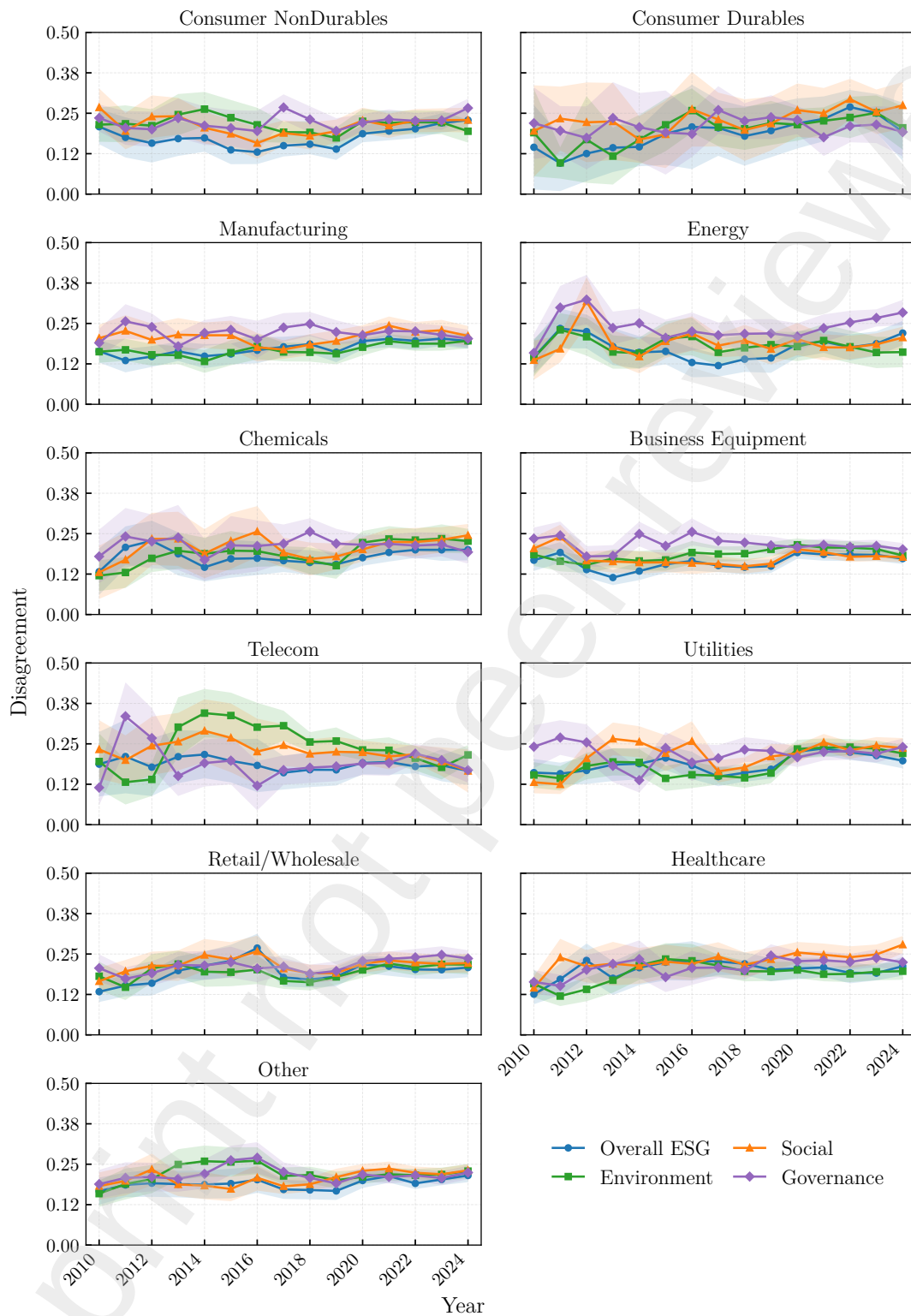
Hyperparameter	Description	Search Distribution
<i>Architecture</i>		
hidden_layer_sizes	Hidden layer dimensions	1–4 layers, 3–48 units
activation	Activation function	{tanh, relu}
<i>Optimization</i>		
solver	Optimization algorithm	{lbfgs, adam, sgd}
learning_rate_init	Initial learning rate	$\log \mathcal{U}(5 \times 10^{-5}, 0.05)$
learning_rate	Learning rate schedule	{adaptive, invscaling, constant}
batch_size	Mini-batch size	{8, 24, 48, 96}
max_iter	Maximum iterations	{75, 125, 175, 250, 350, 450, 600}
tol	Convergence tolerance	$\log \mathcal{U}(10^{-6}, 10^{-2})$
<i>Regularization</i>		
λ_t	L2 penalty (weight decay)	$\log \mathcal{U}(10^{-4}, 5)$
early_stopping	Early stopping enabled	{True, False}
validation_fraction	Validation set fraction	$\mathcal{U}(0.08, 0.33)$
n_iter_no_change	Patience (epochs)	{3, 7, 12, 18, 25}
<i>Solver-Specific (SGD)</i>		
momentum	Momentum coefficient	$\mathcal{U}(0.70, 0.99)$
nesterovs_momentum	Nesterov momentum	{True, False}
power_t	Inverse scaling exponent	$\mathcal{U}(0.3, 0.9)$
<i>Solver-Specific (Adam)</i>		
beta_1	First moment decay	$\mathcal{U}(0.80, 0.99)$
beta_2	Second moment decay	$\mathcal{U}(0.95, 0.999)$
epsilon	Numerical stability	$\log \mathcal{U}(10^{-10}, 10^{-5})$

Notes: Hyperparameters optimized via randomized search with 50 configurations at each formation date. $\mathcal{U}(a, b)$ denotes uniform distribution over $[a, b]$; $\log \mathcal{U}(a, b)$ denotes log-uniform distribution; $\{\cdot\}$ indicates discrete sampling. SGD and Adam parameters apply only when the corresponding solver is selected.

Figure A.1: TIME SERIES OF ESG RATINGS AND DISAGREEMENT BY INDUSTRY



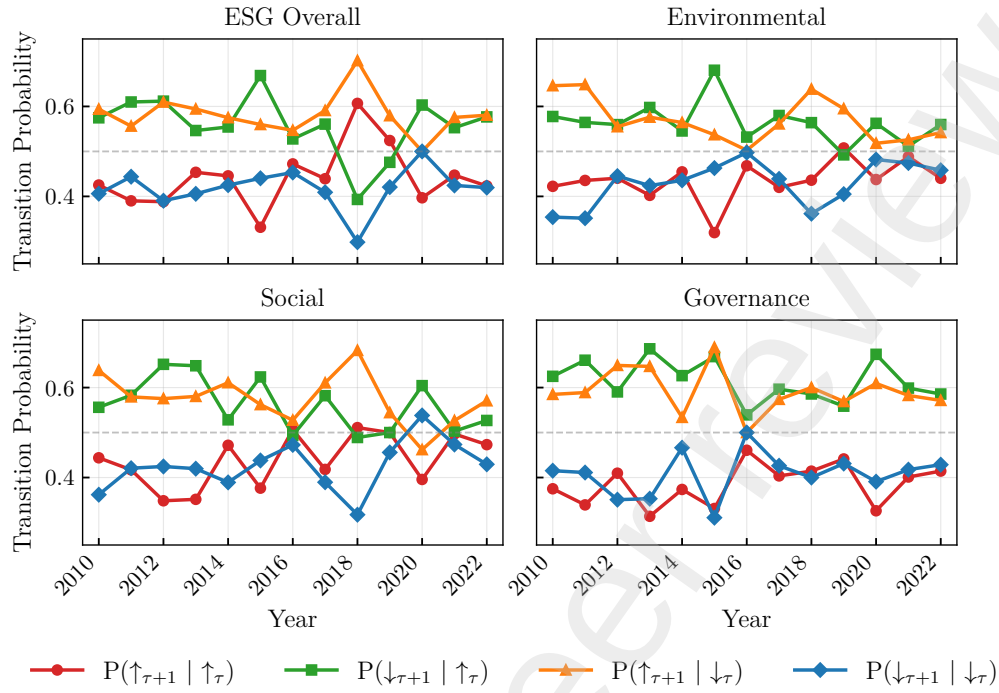
(B) ESG RATING DISAGREEMENT BY INDUSTRY



Notes: Each panel displays one Fama–French 12 industry (excluding Finance) for S&P 500 firms from 2010 to 2023. Panel (a) plots the cross-sectional mean of average ESG percentile ranks across available rating providers. Panel (b) plots mean pairwise disagreement as defined in Equation 3. Shaded regions indicate 95% confidence intervals around the cross-sectional mean.

Appendix B. Additional Results

Figure B.1: TRANSITION PROBABILITIES OF ESG RATING DISAGREEMENT CHANGES



Notes: Each panel plots four conditional transition probabilities by fiscal year for Overall ESG and the Environmental, Social, and Governance pillars. For each firm i and fiscal year τ , we compute the disagreement change $\Delta D_{i,\tau}$ using balanced pairs as defined in Equation 4. We define the binary direction indicator $S_{i,\tau} = \uparrow$ if $\Delta D_{i,\tau} > 0$ (disagreement increased) and $S_{i,\tau} = \downarrow$ if $\Delta D_{i,\tau} \leq 0$ (disagreement decreased or unchanged). For each year τ , transition probabilities are computed as: $P(\uparrow_{\tau+1} | \uparrow_{\tau}) = N_{\uparrow\uparrow}/N_{\uparrow}$ where $N_{\uparrow\uparrow}$ is the count of firms with $S_{i,\tau} = \uparrow$ and $S_{i,\tau+1} = \uparrow$, and $N_{\uparrow}$ is the count of firms with $S_{i,\tau} = \uparrow$. The other three probabilities are defined analogously. Transitions require consecutive fiscal years. $P(\uparrow | \uparrow)$ (red circles) and $P(\downarrow | \downarrow)$ (blue diamonds) measure persistence; $P(\downarrow | \uparrow)$ (green squares) and $P(\uparrow | \downarrow)$ (orange triangles) measure reversal. The dashed line at 50% indicates no serial dependence.

Figure B.2: TRANSITION PROBABILITIES OF ESG RATING DISAGREEMENT BY INDUSTRY



Notes: Each panel displays average transition probabilities across the Fama-French 12-industry classification (excluding Finance) for Overall ESG and the Environmental, Social, and Governance pillars. The binary direction indicator $S_{i,\tau} \in \{\uparrow, \downarrow\}$ equals $\uparrow$ if disagreement increased from fiscal year τ to $\tau + 1$ and $\downarrow$ otherwise, computed using balanced pairs only (provider pairs with coverage in both years). Columns represent four conditional transitions: $\uparrow | \uparrow$ is $P(S_{i,\tau+1} = \uparrow | S_{i,\tau} = \uparrow)$, the probability of continued increase; $\downarrow | \uparrow$ is $P(S_{i,\tau+1} = \downarrow | S_{i,\tau} = \uparrow)$, reversal after increase; $\uparrow | \downarrow$ is $P(S_{i,\tau+1} = \uparrow | S_{i,\tau} = \downarrow)$, reversal after decrease; $\downarrow | \downarrow$ is $P(S_{i,\tau+1} = \downarrow | S_{i,\tau} = \downarrow)$, continued decrease. Cell values show probabilities pooled across fiscal years; colors range from green (lower) to red (higher). Industry abbreviations: NoDur (Consumer Nondurables), Durbl (Consumer Durables), Manuf (Manufacturing), Enrgy (Energy), Chems (Chemicals), BusEq (Business Equipment), Telcm (Telecom), Utils (Utilities), Shops (Retail/Wholesale), Hlth (Healthcare), Other.

Table B.1: OUT-OF-SAMPLE CLASSIFICATION PERFORMANCE: LOGISTIC MODELS

Training Period	Test Year	Overall ESG	Environmental	Social	Governance
2010–2016	2017	0.584	0.624	0.630	0.605
2010–2017	2018	0.625	0.635	0.633	0.585
2010–2018	2019	0.649	0.627	0.652	0.647
2010–2019	2020	0.591	0.606	0.585	0.584
2010–2020	2021	0.613	0.592	0.634	0.573
2010–2021	2022	0.564	0.540	0.546	0.573
2010–2022	2023	0.612	0.623	0.601	0.583
Mean		0.605	0.607	0.612	0.593

Notes: Out-of-sample AUC for the logistic model predicting whether firm-level ESG rating disagreement increases over the subsequent fiscal year. The logistic models use identical features, preprocessing, and walk-forward cross-validation as the MLP reported in Table 2. AUC = 0.5 indicates no discriminative ability beyond random classification; AUC = 1.0 indicates perfect separation.

Table B.2: RETURN PREDICTABILITY: LOGISTIC MODEL

	Overall ESG			Environmental			Social			Governance		
	F	E	L	F	E	L	F	E	L	F	E	L
Panel A: Main Variable of Interest												
$\hat{p}_{i,t}$	0.055 (0.62)	0.312 (0.32)	-0.050 (-0.48)	0.184 (1.32)	-0.054 (-0.32)	0.438* (1.65)	0.035 (0.33)	-0.099 (-0.65)	0.147 (0.89)	-0.105 (-0.72)	-0.223 (-1.04)	-0.093 (-0.46)
Panel B: Firm Characteristics												
Size	0.001 (0.19)	0.016* (1.69)	-0.004 (-0.36)	0.002 (0.21)	0.018* (1.80)	-0.004 (-0.36)	-0.000 (-0.04)	0.016* (1.73)	-0.006 (-0.53)	0.001 (0.14)	0.017** (2.01)	-0.006 (-0.51)
B/M	-0.012** (-1.98)	-0.030*** (-3.07)	-0.003 (-0.35)	-0.013** (-2.01)	-0.030*** (-3.07)	-0.004 (-0.43)	-0.013** (-2.02)	-0.030*** (-3.13)	-0.004 (-0.38)	-0.012* (-1.94)	-0.030*** (-3.14)	-0.004 (-0.37)
Prof	0.057 (1.38)	0.112** (2.28)	-0.024 (-0.35)	0.053 (1.26)	0.111** (2.27)	-0.037 (-0.54)	0.053 (1.27)	0.111** (2.24)	-0.030 (-0.44)	0.058 (1.38)	0.112** (2.29)	-0.027 (-0.39)
Mom	-0.071*** (-3.41)	-0.002 (-0.08)	-0.143*** (-5.07)	-0.071*** (-3.46)	-0.002 (-0.07)	-0.147*** (-5.36)	-0.071*** (-3.42)	-0.004 (-0.13)	-0.143*** (-5.06)	-0.070*** (-3.37)	-0.001 (-0.02)	-0.143*** (-5.06)
Disp	0.007*** (2.88)	-0.077 (-1.29)	0.006*** (2.93)	0.006*** (2.66)	-0.078 (-1.30)	0.005*** (2.58)	0.007*** (3.00)	-0.076 (-1.27)	0.006*** (2.93)	0.007*** (2.89)	-0.075 (-1.27)	0.007*** (3.17)
Beta	0.011 (1.18)	-0.002 (-0.37)	0.041 (1.42)	0.011 (1.18)	-0.002 (-0.34)	0.039 (1.36)	0.012 (1.21)	-0.002 (-0.35)	0.042 (1.48)	0.012 (1.18)	-0.003 (-0.41)	0.041 (1.40)
Vol	0.004 (0.45)	-0.007 (-0.51)	0.016 (1.00)	0.004 (0.43)	-0.007 (-0.50)	0.014 (0.88)	0.004 (0.46)	-0.007 (-0.47)	0.016 (0.99)	0.004 (0.43)	-0.007 (-0.50)	0.015 (0.93)
Panel C: ESG Variables												
D Level	-0.034 (-0.41)	-0.002 (-0.03)	-0.188 (-1.33)	0.114 (0.68)	-0.032 (-0.17)	0.320 (0.93)	-0.084 (-0.83)	-0.127 (-1.08)	-0.021 (-0.13)	-0.032 (-0.51)	0.022 (0.28)	-0.113 (-1.14)
ESG Level	-0.004 (-0.15)	0.020 (0.53)	-0.031 (-0.66)	-0.042 (-1.32)	0.006 (0.13)	-0.095* (-1.66)	0.024 (0.83)	0.029 (0.73)	-0.005 (-0.11)	0.011 (0.31)	0.000 (0.01)	0.014 (0.27)
Model Statistics												
N	2,278	981	1,297	2,278	981	1,297	2,278	981	1,297	2,278	981	1,297
Adj. R^2 (%)	24.244	11.911	31.965	24.378	11.889	32.374	24.325	12.022	31.996	24.231	11.974	31.886
Ind + Year FE												✓

Notes: This table reports panel regressions of annual stock returns on the predicted probability of an ESG disagreement increase from the logistic model. The dependent variable is the buy-and-hold return from July of year t through June of year $t + 1$. $\hat{p}_{i,t}$ is the predicted probability of pillar-specific disagreement increase from the logistic model, estimated using only information available at portfolio formation. F denotes the full out-of-sample period (2017–2023); E denotes the early period (2017–2019); L denotes the late period (2020–2023). Size is the natural log of market capitalization. B/M is the natural log of the book-to-market ratio. Prof is revenues minus cost of goods sold, scaled by total assets. Mom is the cumulative return over the 11 months ending in May of formation year t , skipping the month immediately prior to formation (June of year t). Disp is the standard deviation of analyst EPS forecasts scaled by the absolute value of mean forecast. Beta and Vol are estimated over 12 months prior to portfolio formation. All pillar-specific specifications use corresponding pillar measures: D Level denotes the baseline disagreement level for that pillar, and ESG Level denotes the average rating level for that pillar. All specifications include industry and formation-year fixed effects. t -statistics based on standard errors clustered at the firm level are reported in parentheses. Bold indicates significance at the 5% level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table B.3: PORTFOLIO PERFORMANCE: LOGISTIC MODEL

	Overall ESG	Environmental	Social	Governance
Panel A: Equal-Weighted Portfolios				
<i>Low Portfolio</i>				
Mean Return	0.89** (2.18)	0.84** (2.12)	0.80* (1.83)	0.86** (1.98)
Sharpe Ratio	0.16	0.16	0.15	0.16
FF6 α	-0.10 (-0.84)	-0.18 (-1.43)	-0.21** (-2.00)	-0.17 (-1.37)
<i>Mid Portfolio</i>				
Mean Return	0.89** (2.08)	0.92** (2.09)	0.95*** (2.63)	0.96*** (2.62)
Sharpe Ratio	0.16	0.16	0.18	0.18
FF6 α	-0.14 (-1.33)	-0.13 (-1.49)	-0.04 (-0.34)	-0.13 (-1.03)
<i>High Portfolio</i>				
Mean Return	1.03*** (2.59)	1.05*** (2.67)	1.05** (2.44)	0.98** (2.21)
Sharpe Ratio	0.19	0.20	0.19	0.17
FF6 α	-0.05 (-0.41)	0.03 (0.26)	-0.04 (-0.36)	0.02 (0.14)
High-Low Spread				
Mean Return	0.14 (1.25)	0.21** (2.49)	0.25*** (3.29)	0.12 (0.72)
Sharpe Ratio	0.11	0.21	0.25	0.07
FF6 α	0.05 (0.44)	0.21** (2.42)	0.17** (2.19)	0.19 (1.44)
Panel B: Value-Weighted Portfolios				
<i>Low Portfolio</i>				
Mean Return	1.11*** (2.84)	1.03** (2.21)	1.11*** (2.98)	1.04*** (2.75)
Sharpe Ratio	0.22	0.19	0.22	0.22
FF6 α	0.06 (0.68)	0.07 (0.56)	-0.01 (-0.09)	-0.03 (-0.19)
<i>Mid Portfolio</i>				
Mean Return	1.14*** (2.60)	1.29*** (3.20)	1.32*** (3.37)	1.58*** (3.89)
Sharpe Ratio	0.21	0.24	0.27	0.31
FF6 α	0.14 (0.75)	0.17 (1.52)	0.23* (1.85)	0.35*** (3.09)
<i>High Portfolio</i>				
Mean Return	1.36*** (2.85)	1.30*** (3.33)	1.26*** (2.67)	1.07** (2.16)
Sharpe Ratio	0.26	0.27	0.23	0.19
FF6 α	0.14 (0.79)	0.10 (0.82)	0.17 (1.46)	0.06 (0.40)
High-Low Spread				
Mean Return	0.25 (0.79)	0.27 (1.60)	0.16 (0.80)	0.03 (0.12)
Sharpe Ratio	0.11	0.11	0.08	0.01
FF6 α	0.08 (0.42)	0.03 (0.32)	0.18 (1.41)	0.09 (0.45)

Notes: This table reports portfolio performance using predicted probabilities from the logistic model. Each year, firms are sorted into terciles based on the predicted probability of a pillar-specific disagreement increase for four dimensions (Overall ESG, Environmental, Social, Governance). For each portfolio (Low, Mid, High, H-L), we report: mean monthly excess return (%), Sharpe ratio, and Fama–French 6-factor alpha. Panel A shows equal-weighted portfolios; Panel B shows value-weighted portfolios. Returns and alphas in % per month. Newey and West (1987) *t*-statistics (12 lags) in parentheses. Bold indicates significance at 5%. ***, **, * denote 1%, 5%, 10% significance.